

Land Restitution, Political Capture and the Fiscal Zombie in Rural Romania

Submitted in Partial Fulfillment of the Requirements for the Degree of
Master of Research in Economics

Sciences Po Paris, School of Research
Department of Economics

Titas Povilaitis

Under the Supervision of

Julia Cagé

Jury Member: **Ekaterina Zhuravskaya**

May 2026

Land Restitution, Political Capture and the Fiscal Zombie in Rural Romania*

By TITAS POVILAITIS[†]

May 2026

Abstract

This paper investigates the long-term economic and political consequences of Romania's 1991 land restitution and its interaction with the 2007 European Union accession shock. Restitution-as-fragmentation plausibly generated a mismatch between new smallholder property rights and the indivisible, large-scale socialist capital stock inherited from collectivization. Using baseline and synthetic difference-in-differences frameworks identified through terrain ruggedness, the analysis finds that this geographic asset mismatch is associated with a substantial decline in mechanized capital across the historically fertile Plains. Post-2007 recovery appears shaped by local institutions: politically competitive Plains adapted through expanding rural enterprise, while clientelist networks in captured counties are linked to weaker commercial adaptation, with business formation approximately half that of uncaptured Plains. Politically controlled counties also show a 14% relative increase in fiscal dependency, consistent with a Fiscal Zombie equilibrium. In bridging spatial economics with institutional theory, the paper suggests that geography may bound initial capacities, while clientelists instrumentalize these vulnerabilities. (JEL P26, N54, R11, Q15, D72, H77)

*I am deeply grateful to my supervisor, Professor Julia Cagé, for her invaluable guidance and for enriching my academic journey throughout this thesis. I also want to thank Professor Ekaterina Zhuravskaya for her advice and for serving on my jury, and to Professor Paul Bouscasse for his instrumental help in shaping the initial concept of this research. My historical analysis of collectivization greatly benefited from the advice and archival expertise of Professors Gail Kligman and Katherine Verdery, as well as from Professor Dorin Dobrinu, whose feedback and suggested literature significantly deepened this project. I am also thankful to Valentina Marican at the National Institute of Statistics of Romania for her crucial assistance in locating essential agricultural data. Of course, I want to thank Prachi, Michelle, and my other friends for their unwavering support and encouraging discussions during this Master's program. Finally, my deepest gratitude goes to my parents — without them, I would not be where I am today. All errors are my own.

[†]titas.povilaitis@sciencespo.fr

Contents

I	Introduction	3
II	Historical and Institutional Background	8
II.A	Pre-Collectivization Agrarian Landscape	8
II.B	Collectivization, 1949-1962	9
II.C	Restitution and EU Accession	11
III	Conceptual Framework	14
IV	Data	16
V	Identification Strategy	20
V.A	Ordinary Least Squares	20
V.B	Baseline Difference-in-Differences and Dynamic Event Study	21
V.C	Split-Sample Difference-in-Differences	22
V.D	Synthetic Difference-in-Differences	23
VI	Results	24
VI.A	Geographic Baseline	24
VI.B	Asset Mismatch Channel	25
VI.C	Political Capture Channel	26
VI.D	Quantifying the Fiscal Zombie	27
VII	Robustness Checks	28
VII.A	Extreme TRI Split	28
VII.B	Extreme PSD Split	29
VII.C	Truncated Timeline	29
VII.D	Leave-One-Out Analysis	30
VIII	Conclusion	31
	References	33
	Figures and Maps	37
	Tables	47
A	Variable Construction and Sources	54
B	Spatial Variation of Key Variables	57
C	Identification Without Controls	61
D	Robustness	69

I. Introduction

It is true that Romania's Southern and Eastern Plains contain some of the country's most fertile agricultural land, historically sustaining grain exports that earned the region its designation as the breadbasket of Eastern Europe. Yet, contemporary evidence inverts this geographic logic. Ionița and Mărcuță (2023) document that the Northeast and Southeast regions, dominated by open plains, record the lowest average wheat yields in the country, while the fragmented, mountainous Center and Southwest that are agronomically less suited to large-scale cultivation, record higher rates. Figure 1 maps the terrain ruggedness underlying this division. The flattest counties, shown in lighter shades, are precisely those whose natural endowment should favor mechanized, industrial-scale agriculture. That they do not is the puzzle this paper investigates.

The underperformance extends beyond crop yields. As Figure 2 illustrates, rural business formation has grown relatively faster in Mountains counties than in Plains counties since the mid-1990s, a divergence that widened further following Romania's 2007 European Union (EU) accession rather than narrowing as market integration might have been expected to encourage. This pattern sits uncomfortably with standard transition models, which predicted that restoring private property rights would rebuild individual incentives and reform agriculture toward market-oriented productivity (Gorton and Davidova, 2004; Swinnen and Rozelle, 2006). The Plains, whose geography and inherited capital stock were arguably best positioned for precisely this restructuring, instead became a locus of relative stagnation.

One candidate explanation is that EU accession financial transfers have stabilized incomes without addressing pre-existing structural weaknesses. Rizov et al. (2013) find evidence that pre-decoupling Common Agricultural Policy (CAP) subsidies may have reduced incentives to reform rather than facilitating it, suggesting that external fiscal support can mask inefficiencies instead of resolving them. Nevertheless, even this account leaves the geographic asymmetry unexplained: why the Plains specifically, and why persistently? The answer proposed here lies in the design of Romania's 1991 land restitution, which returned collectivized land to pre-communist owners in original strip parcels without any consolidation mechanism and without any provision for the collective infrastructure that socialist agriculture had constructed (Verdery, 2003; Cartwright, 2017). What followed was not simply incomplete privatization, but a mismatch between newly fragmented property rights and the indivisible physical capital that made those rights productive. A mismatch whose consequences, this paper argues, were subsequently set in by local political institutions rather than corrected by market forces.

Explaining this puzzle will require the following question to be asked: *to what extent did the design of Romania's 1991 land restitution — through its interaction with the inherited socialist capital stock — generate structural vulnerabilities that the 2007 EU accession shock subsequently entrenched through political capture?* This framing departs from the dominant strand of transition economics, which evaluated restitution primarily by whether privatization occurred, instead of by how the specific design of property rights assignment interacted with the physical form of the capital stock it took over (Frydman et al., 1999). Romania's Land Law returned holdings to pre-collectivization owners in original strip parcels, a choice

influenced by political symbolism and historical justice rather than productive logic (Verdery, 2003; Kligman & Verdery, 2011).

The economic consequences of that choice, however, were conditioned by what collectivization had physically constructed in the intervening decades, such as infrastructure whose productive value was dependent on the scale and vertical integration of the socialist organizational form it served. When that form dissolved, the question of what happened to the capital it left behind remains underexplored. This paper argues that the answer has implications extending well beyond agricultural productivity. Where the asset mismatch was sharpest, that is, in the Plains, households found themselves asset-rich, but cash-poor, with land that provided subsistence, but insufficient liquid income to finance relocation, likely generating the spatially stable, transfer-dependent population that clientelist political arrangements tend to demand to consolidate (Grzymala-Busse, 2008; Robinson & Verdier, 2013). The paper calls this configuration the Fiscal Zombie equilibrium, and investigates whether it can be traced empirically to the interaction of restitution design, inherited capital, and political capture.

Nevertheless, identifying the causal legacy of restitution design needs a source in variation in asset mismatch severity that is presumably unrelated to post-1991 economic trajectories. The paper exploits terrain ruggedness, which is measured through the Riley et al. (1999) Terrain Ruggedness Index (TRI) computed from satellite elevation data across all 42 Romanian counties, as this identification device. Rugged terrain historically constrained the feasibility of collectivization and the specificity of the capital stock it constructed (Kligman & Verdery, 2011). Flatter counties, thus, inherited a more severe asset mismatch upon restitution, while their mountainous counterparts, where collectivization built less capital-intensive infrastructure, faced comparatively lower adjustment costs. Crucially, terrain is reasonably orthogonal to the post-transition political and economic outcomes the paper measures, lending credibility to its use as a geographic treatment proxy.

Furthermore, the analysis utilizes Romania's 2007 EU accession as an exogenous intensifying shock, externally determined and legally uniform across all counties, but whose distributional consequences were conditioned by the pre-existing asset mismatch and political capture structure. The empirical panel excludes the capital Bucharest and the surrounding Ilfov counties as their economic arrangements are more determined by urban agglomeration effects rather than agricultural history. Consequently, the resulting sample covers 40 counties over 1996–2019, with data sourced primarily from the Romanian National Institute of Statistics (INS, 2026).

The estimation proceeds sequentially. Ordinary Least Squares (OLS) specifications establish the geographic baseline associations, a difference-in-differences (DiD) framework exploiting the 2007 shock estimates the average divergence between Plains and Mountains counties, and a split-sample extension isolates the political capture channel by partitioning Plains counties by their historical Social Democratic Party (PSD) vote share. Because the treatment shock occurs simultaneously for all counties in 2007, the design avoids the negative-weighting concerns associated with staggered rollouts documented by de Chaisemartin and D'Haultfœuille (2023). A synthetic difference-in-differences (SDiD) estimator taken from Arkhangelsky et al. (2021) is then applied to the high-political-capture subsample as the most demanding specifi-

cation, constructing an optimal synthetic counterfactual that filters residual pre-trend noise.

The strategy confirms a substantial geographic penalty associated with Plains status. The OLS baseline shows that privatization of agricultural land proceeded roughly 3.40 percentage points more slowly in the Plains counties than in Mountains counties, and tractor density, which is the paper's primary proxy for productive agricultural capital, averaged approximately 1.78 units per 100 arable hectares lower, a gap equivalent to around 68% of the model's sample mean. Notably, privatization rate does not independently predict this capital deficit, suggesting the tractor shortfall reflects the asset mismatch directly rather than through administrative reform delays. The Plains dummy is also linked to a 7.06 percentage points higher PSD vote share, consistent with the hypothesis that geographic vulnerability and political capture tend to co-locate.

EU accession shock of 2007 appears to have deepened instead of corrected these disparities. Exposure to open European markets is associated with a further gap of around -1.56 tractors per 100 arable hectares in Plains counties relative to Mountains, which is approximately equal to 101% of the variable's overall pre-treatment average. At the same time, aggregate rural business formation paradoxically rises. The split-sample results, however, hint that this effect might be masking two divergent institutional trajectories. In Low-PSD Plains counties, the difference in rural firm registrations, compared to the Mountains benchmark, increases by approximately 35% relative to the full sample mean, indicating successful commercial adaptation following market integration. In High-PSD captured counties, the equivalent estimate is lower at around 10% of the sample mean and statistically indistinguishable from zero, pointing toward the possibility that clientelist networks suppressed the non-farm transition, which accession-era market opening might otherwise have motivated. These politically controlled Plains also show a statistically significant relative decline in rural natural increase of roughly 1.80 per 1,000 inhabitants post-accession, while their Low-PSD counterparts remain comparatively more stable, and net migration patterns, despite being statistically noisy, suggest the High-PSD region remained more spatially constrained throughout with considerably more immigration than emigration, which is consistent with the liquidity trap mechanism.

The SDiD estimates on the high-capture subsample lend further support to the Fiscal Zombie interpretation. Fiscal dependency on central transfers increases by approximately 1.83 percentage points in High-PSD Plains relative to their synthetic Mountains counterfactual, representing a relative increase of around 14% over the pre-treatment mean. A parallel falsification exercise on Low-PSD Plains yields no comparable fiscal effect. This strengthens the case that the dependency pattern is linked to political capture rather than geography alone. The tractor deficit persists at around -0.99 units in these specifications, rural natural increase relatively declines by a further estimated 1.27 per 1,000 inhabitants, and the noisy net migration patterns hint at a population that is aging and increasingly immobile, which are collectively compatible with a captive, state-dependent electorate rather than a region undergoing structural adjustment.

The core findings hold across four robustness checks. Restricting the sample to the flattest and most rugged quartiles of counties, that is, eliminating transitional geographies, amplifies the

tractor deficit to -2.37 units, aligning with the theory of the asset mismatch being a structural phenomenon instead of a boundary artifact. Partitioning Plains counties by the extremes of political capture produces a dose-response pattern consistent with the main findings, with demographic decay even intensifying in the most captured counties. Truncating the panel to 2000–2019 leaves the headline coefficients materially unchanged, hence, ruling out early-transition volatility as a confound. A leave-one-out analysis iterating the 40 counties confirms that no single county drives the aggregate results, including eastern border counties, where post-2007 Romanian passport acquisition by Moldovans risks introducing potential migration measurement noise.

The identification strategy, its aforementioned results, and the paper in general speak most directly to the transition economics literature on privatization design and agricultural restructuring. Frydman et al. (1999) establish that the method of privatization, not merely its occurrence, determines whether the productive restructuring follows, which is a finding that motivates this paper's focus on restitution design rather than privatization rates alone. Where Mathijs and Swinnen (1998) document that regions entering transition with more capital-intensive agricultural systems faced larger adjustment costs, and where Gow and Swinnen (1998) identify the hold-up problem generated when land and processing infrastructure were privatized separately in Romanian transition, the thesis traces what neither does: how that initial hold-up translated into a durable political equilibrium instead of dissipating through market adjustment. Sabates-Wheeler (2002) and Hartvigsen (2014) present fragmentation as a barrier to land market development across Eastern Europe, while Adamopoulos and Restuccia (2020) as well as Foster and Rosenzweig (2022) quantify the scale penalties that smallholder proliferation imposes on mechanized agriculture. This paper's first contribution is to show that fragmentation alone could understate the problem. It was plausibly the interaction between broken up ownership and indivisible infrastructure that rendered the Plains' capital stock effectively worthless, an outcome the restitution-as-fragmentation design made structurally likely from the outset.

Theoretical mechanism linking restitution design to capital destruction draws on transaction cost economics and asset-based poverty trap theory. Klein et al. (1978), Grossman and Hart (1986), and Williamson (1987) show that when assets are highly specific to a particular governance structure, dissolving that composition collapses asset value to a negligible next-best use. It is exactly this dynamic the paper documents for Plains machinery pools, such as the tractors, following collectivization's dismantling. De Soto (2001) offers the dead capital concept as a description of this outcome, while Besley (1995) as well as Deininger and Jin (2006) establish that insecure property rights further suppress productive investment by raising its perceived risk.

The resulting liquidity constraint maps onto Zimmerman and Carter (2003) as well as Carter and Barrett (2006), who theorize that households below a critical asset threshold become locked in low-level equilibrium. The closest structural comparator here is Bleakley and Ferrie (2014), who find that an exogenous land windfall on the Georgia frontier generated no lasting productivity gain possibly because complementary capital was absent. The Romanian restitution shock operates through a similar mechanism, though in reverse. Formal ownership was restored, while the infrastructure giving that ownership productive value was simultane-

ously deprecated. Manuelli and Seshadri (2014) further suggest that tractor adoption is not automatic, but contingent on relative factor prices, which could help explain why the Plains tractor deficit persisted even as the broader economy modernized. The paper's second contribution, hence, lies in connecting these mechanisms into a coherent account of how physical capital specificity can constitute a persistence channel as binding as the institutional and legal pathways emphasized elsewhere in the literature.

Relevant persistence literature, anchored by Acemoglu et al. (2001), Banerjee and Iyer (2005), Nunn (2008), Dell (2010), as well as Heldring et al. (2015), establishes that historical shocks generate long-run development gaps through cultural and legal channels. In Eastern Europe specifically, Pop-Eleches (2007) as well as Grosfeld and Zhuravskaya (2015) show that pre-industrial imperial boundaries continue to predict voting behavior, while Markevich and Zhuravskaya (2018) demonstrate that, where land following serfdom abolition was assigned without resolving the governance of collective infrastructure, agricultural productivity remained suppressed for decades, which is a parallel whose precision motivates this paper's Romanian application. The thesis extends Pop-Eleches (2007) from institutional to aforementioned physical capital as a persistence channel, showing that the technological legacy of 1950s collectivization presumably continues to shape the political behavior in the 2000s.

The final contribution concerns the political consequences of the asset-based immobility trap. Hirschman (1970) provides the foundational logic. Where exit is unavailable, voice becomes the only disciplining mechanism, and where voice is itself captured, political lock-in follows. Iyer and Mani (2012) show empirically that bureaucrats exercise greater extractive control over immobile citizens, while O'Dwyer (2006) and Grzymala-Busse (2008) document how political parties essentially colonize public administration to sustain patronage rather than service delivery. Stokes et al. (2013) as well as Mares and Young (2019) provide the micro-foundations. Brokers target immobile, welfare-dependent voters with conditional transfers, locking dependent populations into stable clientelist equilibria, consistent with Robinson and Verdier (2013) who show that clientelism tends to emerge as an optimal strategy exactly in low-productivity environments.

On the migration side, Sandu (2005), Bryan et al. (2014), Angelucci (2015), as well as Dustmann and Görlach (2016) all establish that migration functions as a liquidity-constrained investment, explaining why transnational exit developed faster in Mountains communities with portable capital and formed pioneer networks than in the cash-poor Plains. What none of these papers provides is an account of how a specific property rights design produces the economic conditions that political capture then instrumentalizes into a stable equilibrium. This paper's third contribution is precisely that account, which is to show that, while geography may have bounded the initial physical shock, the so-called Fiscal Zombie equilibrium that followed was plausibly the product of political choice rather than geographic inevitability.

Taken together, the three mechanisms — asset mismatch, immobility trap, and political capture — form a cohesive causal chain linking a specific restitution design choice in 1991 to observable patterns of fiscal dependency and demographic decay in the 2000s. Each link in this sequence is examined one after another, allowing the analysis to distinguish what geog-

raphy determined from what political institutions subsequently shaped. The remainder of the paper is organized as follows. [Section II](#) provides the historical and institutional background, covering the context of pre-collectivization, the 1949–1962 collectivization campaign, and the restitution legislation together with the EU accession shock. [Section III](#) develops the conceptual framework. [Section IV](#) describes the data and variable construction, whereas [Section V](#) presents the identification strategy. [Section VI](#) and [Section VII](#) report the main results and robustness checks, respectively, and [Section VIII](#) concludes.

II. Historical and Institutional Background

A. Pre-Collectivization Agrarian Landscape

Romania’s agriculture on the eve of collectivization was already defined by geographic duality that the socialist transformation would later exploit and ultimately entrench in the post-communist period. The interwar land reforms of 1921, which redistributed approximately 6 million hectares from large estates to peasant smallholders, had produced a formally egalitarian, but structurally fragmented agricultural sector.¹ Yet this fragmentation manifested differently across terrain. In the Southern and Eastern Plains, which were the historical Bessarabia and Old Kingdom lowlands, pre-reform estates had been larger, relatively more mechanized, and oriented toward grain monoculture for export, meaning that post-reform fragmentation produced holdings that were individually too small to sustain the capital-intensive farming the terrain demanded. Gusti’s (1938) *Enciclopedia României* records average smallholder (*improprietăriți*) holdings as of 1934 at approximately three hectares in the Plains regions (Old Kingdom and Bessarabia) and just one hectare in the Mountains (Transylvania and Bucovina at the time). This is a distribution that already signals a limitation of holding size alone as an explanatory variable for post-transition outcomes since Mountains smallholders operated on smaller plots yet adapted more successfully after 1991 (Gusti, 1938).² At the other end of the distribution, large landowners averaged approximately 400 hectares in the Plains against 140 hectares in the Mountains, which is a scale differential that had possibly already generated an industrial logic in Plains agriculture, with large estates requiring mechanized equipment, irrigation infrastructure, and vertically integrated processing arrangements that smallholder Mountains farming had developed relatively less (Adamopoulos & Restuccia, 2020; Foster & Rosenzweig, 2022).

In the Mountains, by contrast, the pre-collectivization system was built around mixed, low-capital agriculture, such as livestock rearing, orchard cultivation, small-scale dairy, whose productive logic was household-scale and whose capital stock was correspondingly portable and divisible. A family’s livestock, its fruit trees, its small plot of maize did not require much vertical integration with processing infrastructure to retain productive value. In the Plains,

¹The 1921 reform was the largest land redistribution in interwar Eastern Europe by area. See Roberts, H. L. (1951) *Rumania: Political Problems of an Agrarian State*. New Haven: Yale University Press, pp. 12–18.

²The 1934 data on landholding structure by region remain one of the few pre-communist sources with region-level granularity. Direct digitization of equivalent census-level agricultural holdings tables was confirmed as unavailable by Professors Kligman and Verdery in correspondence with the authors (November 2025), who noted that raw figures would require archival recovery in Romania.

however, even smallholder agriculture operated in proximity to, and partial dependence on, the large-estate infrastructure that dominated the landscape. This asymmetry in the nature of the pre-existing capital stock, which is scale-specific and infrastructurally dependent in the Plains, portable and self-contained in the Mountains, is the baseline condition from which all subsequent divergence flows. Collectivization did not create this duality, but it built a new and more rigid layer of fixed, specific capital on top of it in the Plains, and the manner of its dismantling after 1989 would determine which regions could reconstitute productive agriculture and which could not. This and other significant institutional transformations, starting from the 1949 collectivization decree through the 1991 restitution legislation and the 2007 EU accession shock will not only be discussed in this section, but will also be situated and summarized chronologically in [Figure 3](#).

B. Collectivization, 1949-1962

Romanian agricultural collectivization was initiated by decree in March 1949, following the consolidation of Communist Party control over the state apparatus and the formal adoption of the Soviet model of forced collectivization as official policy.³ The campaign proceeded in three broadly identifiable phases: an initial coercive push between 1949 and 1953, a tactical retrenchment following Stalin's death and the 1953 New Course, and a final accelerated drive between 1957 and 1962 that achieved formal completion.⁴ By June 1962, the Romanian state announced that 94% of arable land had been incorporated into collective agricultural units (*Cooperative Agricole de Producție*, CAP), based on nominally pooled peasant land, or state farms (*Întreprinderi Agricole de Stat*, IAS) operated with wage labor, making Romania's collectivization one of the most complete in the Eastern Bloc outside the Soviet Union itself (Kligman & Verdery, 2011).

The institutional mechanism of collectivization combined legal coercion with direct physical violence. Peasants were formally required to sign voluntary adhesion forms joining the CAP, but the voluntariness of this process was almost completely undermined by the secret police (*Securitate*) surveillance, arbitrary detention, confiscation of livestock and equipment, as well as the threat of criminal prosecution under the charge of sabotage of socialist construction (Iordachi & Dobrinicu, 2009). The *chiaburi* — wealthy peasants defined loosely by landholding size and perceived class status — were subjected to disproportionate repression as the ideologically designated class enemy of collectivization, facing not only disproportionate arrest rates, but also extensive exclusion from CAP membership, effectively expropriating their assets without any compensatory mechanism.⁵

³The Romanian Workers' Party Plenum of March 3–5, 1949 formally launched the collectivization campaign. See Iordachi & Dobrinicu (2009), p. 27.

⁴Borșa (2013) provides a concise periodization of the campaign phases and their alignment with Soviet directives, noting that Romanian authorities adopted the Soviet model with minimal institutional adaptation.

⁵The legal definition of *chiabur* was purposefully kept imprecise, allowing local party activists to apply it opportunistically. See Kligman and Verdery (2011), pp. 113–119, for an analysis of how this definitional flexibility served coercive ends.

The geographic variation in collectivization intensity is directly relevant to this paper. As Kligman and Verdery (2011) emphasize, the speed and intensity of the campaign varied considerably across regions, mediated by the interaction of local social structures, party activist constraints, and geo-historical conditions.⁶ While Figures 4 and 5 display the speed and intensity variation across geography, Panels A and B of Table 1 quantify it in collectivization rates (Kligman & Verdery, 2011) as well as arrests, which specifically use data from Ivănescu (2015), who recovered detention figures from the *Securitate* archives (SRI, 1994) for the peak repression year of 1959 in 16 regions at the time.⁷ Here, the magnitude of the arrests in region r is estimated in this thesis with an index made using the following formula:

$$\text{ArrestIndex}_r = \frac{\text{PeasantsArrests}_r}{\text{RuralPopulation}_r} \times 10,000$$

weighting arrests per 10,000 rural inhabitants using population figures from the 1956 Romanian census (Suret-Canale, 1956). A parallel index is calculated for *chiaburi* arrests. The Plains regions record a mean collectivization rate of 57.4% by 1960, against 29.8% in the Mountains regions, a difference of nearly 28 percentage points, which, despite inflated standard deviation due to a small sample size, suggests that terrain ruggedness could be an eligible proxy for collectivization speed and intensity.

The arrest data, however, produce a counterintuitive pattern that merits careful interpretation. Mountains regions record a higher mean Peasants Arrests Index (74.70 versus 26.90 for the Plains) and a substantially higher *chiaburi* index (63.00 versus 11.10). This does not seem to indicate that collectivization was more intense in the Mountains in any productive sense and, instead, it plausibly reflects the greater resistance capacity of Mountains communities, whose terrain afforded both physical refuge and stronger pre-existing social cohesion, motivating the state to deploy more coercive resources to achieve a lower final collectivization rate (Kligman & Verdery, 2011). In the Plains, the flat, exposed terrain and the legacy of large estate agriculture, which had probably already eroded smallholder solidarity, meant that resistance was structurally weaker and collectivization could be achieved at lower arrest intensity, but higher ultimate coverage. The standard deviation on both arrest indices is extremely high, reflecting genuine regional heterogeneity rather than a uniform repressive pattern, which is a finding consistent with the regional variability emphasized by Kligman, Verdery, and Dobrinu.

What collectivization ultimately produced in the Plains was not only a change in ownership structure, but the construction of a new physical infrastructure calibrated to industrial-scale agriculture: large-scale irrigation systems covering hundreds of thousands of hectares, centralized grain storage facilities, mechanized equipment pools, and processing plants integrated

⁶Professor Gail Kligman, in correspondence with the author (November 2025), confirmed that this regional variability is a central analytical concern of Kligman and Verdery (2011) and also directed the author to Iordachi and Dobrinu (2009) for detailed case study evidence on individual regions.

⁷Professor Dorin Dobrinu, in correspondence with the author (October 2025), confirmed that thorough county-level arrest data remain fragmentary and that Ivănescu's *Securitate*-based figures (SRI, 1994) represent the most complete available source at the regional level. Dobrinu directed the author to his own documentary work for supplementary evidence: Dobrinu (2004).

vertically within the CAP and IAS organizational structure.⁸ This infrastructure was likely not incidental to the collective farm, it was rather constitutive of it, and its productive value was predominantly conditional on the scale and administrative integration of the socialist organizational form. The implications of this specificity for the post-1991 transition are developed later in this section.

Between 1949 and 1989, the Romanian state directed considerable capital investment into agricultural infrastructure in the Plains, concentrated in such areas as mechanization, irrigation, centralized storage, and processing capacity (Cartwright, 2017). Romania had effectively no domestic tractor production at the end of the Second World War except when the Braşov factory was converted from aircraft to tractor manufacturing only under Soviet occupation. The subsequent expansion of the national tractor fleet from near zero to industrial scale within three decades was almost entirely absorbed by the CAP and IAS units of the Southern and Eastern Plains, where large-scale grain monoculture required mechanized tillage and harvesting at a scale impossible for individual smallholders.⁹ Storage capacity was also centralized. Each CAP operated collective grain depots and processing facilities were integrated directly into state procurement chains, bypassing market mechanisms entirely.

In the Mountains, the collectivization period did not produce as much equivalent capital accumulation. The terrain precluded large-scale irrigation since the CAP units established there were smaller, less mechanized, and organized around livestock rather than grain, meaning the capital stock they operated, such as barns, small equipment, pasture management infrastructure, was closer in character to what had existed before collectivization than to the industrial apparatus constructed in the Plains. Consequently, the aforementioned Mathijs and Swinnen (1998) document this contrast thoroughly across Central and Eastern European transition economies and find that regions entering the transition with more large-scale, irrigation-dependent agricultural systems would experience structurally larger costs of adaptation precisely because their inherited assets were less redeployable at the household scale.

C. Restitution and EU Accession

The infrastructure was inherited by the Romanian state in December 1989, but the legal framework designed to dismantle socialist agriculture effectively treated it as though it did not exist. **Law 18/1991**, the foundational land restitution legislation passed by the Romanian parliament on February 19, 1991, established the formal mechanism for returning CAP-collectivized land to its pre-communist owners or their heirs, generally bypassing the IASs.¹⁰ The law mandated restitution of up to ten hectares per family in the original plot configurations, that is, in the

⁸By 1989, Romania's irrigation infrastructure covered approximately 3 million hectares, the vast majority concentrated in the Southern and Eastern Plains. See Urşanu et al. (2024), available at: <https://www.sciencedirect.com/science/article/pii/S2213305424000262>.

⁹On the Braşov factory conversion and the origins of Romanian tractor production under Soviet occupation, see Deletant, D. (2019) *Romania under Communism: Paradox and Degeneration*. London: Routledge, pp. 89–92.

¹⁰The official name of the law being *Legea fondului funciar nr. 18/1991*, it was published in *Monitorul Oficial al României, Partea I, nr. 37*, February 20, 1991. The full text (in Romanian) is available via the parliament's legislative database at <https://legislatie.just.ro/Public/DetaliiDocument/280373>.

strip parcels that had existed prior to collectivization, essentially without any consolidation mechanism, without any provision for the collective management of inherited infrastructure, and without any compensation framework for assets that could not be meaningfully individualized.¹¹ Irrigation canals, machinery depots, storage silos, and processing facilities, which are the indivisible collective infrastructure documented in the previous paragraphs, fell outside the restitution framework, passing nominally to local authorities or newly formed commercial companies, but without the financial resources, technical capacity, or legal clarity required to maintain them (Verdery, 2003).

The consequences were immediate. Without a functioning operator for the irrigation infrastructure, maintenance almost fully ceased and the network deteriorated rapidly. World Bank estimates from the mid-1990s suggest that effective irrigated area in Romania fell from approximately 3 million hectares at the end of the communist period to under 1 million hectares by 1995, a collapse of more than two thirds within five years of restitution.¹² Machinery pools were either auctioned at distressed prices to private buyers who often lacked the scale to deploy them productively, or simply abandoned. The result was the dead capital configuration discussed in the introduction (De Soto, 2001) — assets with significant replacement value, but negligible productive value in the new property rights environment.

Verdery (2003) captures the political logic underlying this design with particular precision. Land restitution in Romania, as mentioned previously, was conceived primarily as a symbolic act of historical justice and a repudiation of communist expropriation, rather than as an economic policy aimed at productive restructuring. The consequence was the aforementioned fuzzy property, which is formal ownership rights that were legally ambiguous, spatially contested, and institutionally unsupported, discouraging both consolidation and investment. Cartwright (2017) reinforces this interpretation, documenting how the restitution commissions established under Law 18/1991 were usually under-resourced, politically manipulated, and technically incapable of resolving the competing claims that strip restitution inevitably generated.¹³

Subsequent legislative attempts to address these failures were partial and belated. **Law 169/1997** amended the restitution framework to extend coverage and correct some procedural irregularities, and **Law 1/2000** further expanded restitution to include forest land and raised the ceiling to fifty hectares. Neither law introduced a consolidation mechanism or resolved the governance vacuum over collective infrastructure. The fundamental asset mismatch, which entails fragmented private land ownership coexisting with orphaned collective capital, remained

¹¹Article 8 of Law 18/1991 explicitly stipulated restitution in the original location (*pe vechiul amplasament*) where physically possible, a provision that directly produced strip fragmentation in the Plains where pre-collectivization plots had been narrow and elongated. See Cartwright (2017) for a detailed legal analysis of this provision and its practical consequences.

¹²Such data can be found in the World Bank Integrated Water Resources Rapid Assessment Report of 2013: Romania. Washington D.C.: World Bank. It is available at: <https://documents1.worldbank.org/curated/en/675851468094456585/pdf/842640WP0P14660Box0382136B00PUBLIC0.pdf>.

¹³Cartwright (2017) also documents numerous cases in which restitution commissions awarded overlapping claims to the same plot, leaving ownership legally unresolved for years and effectively preventing any productive use of the land.

structurally intact throughout the 1990s and into the 2000s, and it is this configuration that the empirical analysis of the paper inherits as its baseline condition.

Since the 1991 restitution legislation, Romania's accession to the European Union on January 1, 2007 represented one of the most significant external institutional shocks to the agricultural sector, introducing three simultaneous changes whose distributional consequences across terrain types were far from uniform.

The first and most immediate change was the activation of Common Agricultural Policy direct payments. Romanian farmers became eligible for the Single Area Payment Scheme (SAPS), a simplified per-hectare subsidy disbursed regardless of production decisions.¹⁴ Because SAPS payments are proportional to registered land area rather than to productive output or capital investment, they tended to favor large landholders and corporate farm operators who had consolidated significant acreage in the Plains during the 1990s and not the smallholder households that restitution had created. Rizov et al. (2013) find that pre-decoupling CAP subsidies negatively affected farm total factor productivity across EU member states, suggesting that subsidy access may have reduced the pressure to restructure rather than facilitating it.

The second change was the activation of single market labor mobility rights. Romanian citizens gained the right to work legally across most EU member states from 2007 onward, accelerating an emigration process that had already begun through a visa-free regime started in 2002 as well as through bilateral agreements with Spain and Italy in the early 2000s.¹⁵ The key distributional question is whether accession-era emigration was accessible equally across terrain types. Sandu (2005) establishes that migration from Romania is strongly network-dependent, with transnational migration chains concentrated in communities that had developed pioneer migrants in earlier waves, which were communities located more often in the Mountains and in western regions with stronger pre-existing connections to Western European labor markets. Plains communities, whose spatial immobility had plausibly prevented the accumulation of transnational social capital during the 1990s, faced relatively higher barriers to participating in post-accession labor mobility, although the relationship is heterogeneous rather than deterministic.

Finally, the third change was the requirement to implement EU rural development regulations under a pre-accession instrument known at first as the Special Accession Programme for Agriculture and Rural Development (SAPARD) and subsequently as the European Agricultural Fund for Rural Development (EAFRD) framework. It conditioned structural funds on administrative capacity and co-financing ability that the Plains counties were often less equipped to mobilize despite also receiving subsidies through other EU-financed frameworks.¹⁶

¹⁴Romania adopted SAPS rather than the standard CAP payment entitlement system, a transitional arrangement permitted for new member states under European Council (EC) Regulation No. 1782/2003. SAPS payments were phased in gradually, reaching 100% of the standard rate by 2016.

¹⁵Spain and Italy, the two primary destinations for Romanian labor migrants, opened their labor markets to Romanian workers progressively from 2002 onward through bilateral agreements, predating formal accession. See Sandu (2005) for the pre-accession migration network structure.

¹⁶Council Regulation (EC) No. 1268/1999 established the SAPARD instrument for pre-accession rural development; it was succeeded by the EAFRD under Council Regulation (EC) No. 1698/2005 upon accession.

Naturally, these three simultaneous changes did not operate in a neutral institutional environment. In the Plains, EU accession intersected with an already-consolidated political economy in which ex-communist elites with more access to land and mechanized capital had established clientelist relationships with PSD-controlled local administrations, collectively influencing the generation of bureaucratic barriers and resource hoarding that negatively affected independent business formation. [Figure 6](#), relevant for the upcoming section, schematizes the resulting equilibrium: sub-market land leases, state contracts channeled through political loyalty, and administrative barriers to entry jointly contributed to the crowding out of the independent business sector that accession-era market integration might otherwise have stimulated in politically controlled Plains counties. Hence, taking this institutional context and the three simultaneous transformations into account, the 2007 accession becomes a methodologically attractive shock for identification purposes. It was externally determined, applied uniformly in legal terms across all Romanian counties, and its timing was sufficiently abrupt to generate pre- and post-accession dynamics. Importantly, because the shock operated through channels, such as subsidy access, labor mobility, administrative capacity, whose distributional consequences were generally conditioned by the pre-existing asset mismatch and the political capture structure, it functions as an intensifier of divergence rather than an independent treatment, which is what the event study and DiD designs in [Section V](#) exploit.

III. Conceptual Framework

The institutional shock of 1991 restitution did not affect Romanian households uniformly as its consequences were mediated by the nature of the capital stock that collectivization had constructed, which was itself probably geographically determined by terrain. This section maps the institutional constraints established in [Section II](#) onto the decision environment facing individual actors — peasant households, potential entrepreneurs, the elites, and local politicians — and derives the county-level observables that each actor’s rational response might generate, which are summarized in [Table 2](#).

A Plains household receiving restituted land in 1991 usually faced a constrained choice set. With strip parcels too small for viable mechanized grain cultivation, no access to the irrigation networks or machinery pools that had made parcels productive under the CAP, and insufficient liquid income to finance either productive investment or relocation, the rational response may have been subsistence farming. It was not because households preferred it, but because the asset mismatch mostly eliminated the alternatives. Selling or leasing land was complicated by legally unresolved ownership claims and the absence of functioning land markets. However, migration required upfront resources and the subsistence plots could not generate them. This behavioral lock-in is what eventually could produce the tractor deficit. Plains counties are predicted to exhibit generally decreasing tractor density not because terrain precludes mechanization, but because the hold-up problem eventually destroys the incentive to invest in replacement capital after the vertically integrated CAP structure dissolved (Gow & Swinnen, 1998). Tractors occupy a specific analytical position here. As collective CAP property was disposed of separately from land under Law 18/1991, their post-1991 trajectory is rather determined by market investment decisions, making tractor density a relatively clean measure

of productive restructuring uncomplicated by the legal ambiguities that surround land privatization itself. Plains counties are therefore also predicted to have its tractor deficit widen after 2007 as EU accession accelerated modernization in counties capable of restructuring.

Privatization rate is predicted to be lower in the Plains for the same underlying reason. Where complementary capital decays and property claims are more legally contested, private operation is unattractive and privatization stalls. This should make privatization rate a revealed measure of asset mismatch severity. Thus, it is an outcome in its own right, but also a mediator through which the Plains dummy might operate on some downstream variables. It is included as a right-hand side control in the OLS specifications for remaining outcomes precisely to separate the direct Plains effect from the portion that flows through delayed privatization.

Predictions for rural business formation are conditional rather than unconditional, and this is an important analytical puzzle of the paper. A potential entrepreneur evaluating entry into a Plains county should face two countervailing forces. On the one hand, Plains counties offer better road access, flatter terrain, lower transport costs, and greater market proximity, which are structural advantages that should favor non-farm business formation independently of agricultural performance. On the other hand, in counties where political capture is deeper, the local PSD administration can generate bureaucratic barriers to entry, channel state contracts through political loyalty, and tolerate elite monopolization of mechanized farming and land leasing that raises the effective cost of independent market entry (see [Figure 6](#)). The net prediction could then be the following: unconditionally positive or at least insignificant Plains effect on business formation and a negative or smaller magnitude effect when conditioned on high PSD vote share. Low-PSD Plains counties, not bounded by the capture equilibrium, are predicted to show larger business formation responses to post-2007 market integration.

The fiscal and demographic predictions should follow a similar moderating logic. A local politician governing an immobile, transfer-dependent electorate faces lower costs of clientelist commitment relative to programmatic competition given that the exit option, which would otherwise discipline political actors is almost unavailable. This behavioral incentive may sustain fiscal dependency on central transfers and suppress local revenue generation, but the effect is predicted to be visible only in High-PSD Plains, where the capture equilibrium is operative. Rural natural increase, a variable aiding in mechanically capturing the downstream consequences of selective mobility, is similarly predicted to decline most severely in High-PSD Plains, where economic stagnation is deepest and demographic immobility most entrenched.

However, the second component of the immobility trap, external migration, is a theoretically ambiguous outcome. Pre-2007, the liquidity constraint possibly suppresses migration unconditionally by preventing households from covering relocation costs. Post-accession labor mobility rights reduce this barrier, but heterogeneously since Mountains communities with established transnational pioneer networks may exploit the shock faster, while Plains communities face compounding barriers from both the liquidity constraint and the absence of migration networks ([Sandu, 2005](#)). Hence, the net Plains effect on external migration might just be indeterminate in sign, and the analysis treats it as exploratory, with primary interest in whether post-2007 dynamics differ between High- and Low-PSD Plains rather than in the

unconditional level effect.

IV. Data

Empirical analysis is based on an unbalanced cleaned panel of 40 Romanian counties (*judete*) that originates from the full dataset of 42 counties. However, two counties are excluded from the panel throughout, which are Bucharest and Ilfov. Both are classified as Plains by terrain ruggedness, but their economic structure, fiscal arrangements, and demographic dynamics are largely determined by urban agglomeration effects rather than agricultural history. Including them could introduce an overall comparability problem for all outcome variables whose variation is premised on rural economic structure.

The panel covers the period 1996–2019, with the unit of observation being the county-year. It therefore covers 19 Plains counties and 21 Mountains counties across 24 years, yielding a maximum of 960 county-year observations per variable, with variation in coverage arising from data availability across specific outcomes documented in the descriptive statistics. For example, privatization rate does not contain data post-2014, whereas rural natural increase lacks one observation for Neamț in 1997. Moreover, variables, such as business formation and hospital bed density, have 920 observations due to unavailable data in 1996, which is why only models without controls contain the maximum 960 observations. All economic, demographic, and administrative variables and their components are sourced from the Romanian National Institute of Statistics (INS, *Institutul Național de Statistică*) with the help of its online database and its statisticians.¹⁷ Electoral data are sourced from the Permanent Electoral Authority (AEP, *Autoritatea Electorală Permanentă*).¹⁸

The main treatment variable is a binary Plains dummy constructed from the Terrain Ruggedness Index, calculated directly from a 90-meter resolution digital elevation model sourced from the Shuttle Radar Topography Mission (SRTM, 2008).¹⁹ TRI was computed using the *QGIS* Terrain Analysis plugin, applying the Riley et al. (1999) algorithm, which defines ruggedness as the mean absolute difference in elevation between a focal cell and its eight surrounding neighborhoods. County-level TRI values represent the spatial mean of all raster cells falling within each county boundary. The index, thus, captures local relief variation rather than absolute elevation, which more precisely proxies the flat, irrigable lowland versus rugged, fragmented highland. It is one of the distinctions that historically determined agricultural collectivization suitability — a flatter county is not necessarily low-lying, but it is one whose terrain permitted large-scale mechanized farming. Counties with TRI below the sample median are classified as Plains, whereas those above as Mountains. While the exact TRI means are listed in the Appendix A, Figure 1 already visualized the index spatially, and Figure 7 additionally presents the distribution following the separation by the median. The latter, hence, produces a treatment group of 19 counties (after excluding Bucharest and Ilfov) whose mean

¹⁷I am particularly thankful to Ms. Marican and Mr. Virdol who have assisted in obtaining the crucial datasets on county-level agricultural outcomes. Other variables were sourced from the online database (INS, 2026).

¹⁸Data were compiled with the help of Commit Global platform (AEP, 2026).

¹⁹SRTM elevation data were obtained from the Consortium for Spatial Information database (CGIAR-CSI, 2008).

TRI score is no higher than 26.30, against a control group of 21 Mountains counties whose mean TRI score is 26.50 or more. In order to verify that the TRI distribution constitutes a genuine geographic discontinuity rather than a boundary effect, [Section VII](#) will test this directly in the extreme TRI robustness check.

A central measure of decollectivization intensity, privatization rate, is measured as the percentage of agricultural land privatized over the period 1996–2014, sourced from INS ([2026](#)) land registry records. Here, the 1996–2014 period is the largest possible time window provided as data are no longer reported after 2014. The longer panel is preferred over a shorter early-transition measure because the paper’s interest is the cumulative pace of privatization over the medium run rather than the initial shock. A county that privatized slowly throughout the 1990s and 2000s is one where the governance vacuum over collective infrastructure probably persisted the longest and it is this persistence that the variable is intended to capture. As discussed in [Section III](#), privatization rate serves a dual role in the empirical strategy first as an outcome in the OLS baseline and then as a right-hand side mediator control in the OLS specifications for the other outcomes in order to isolate the Plains effect net of the portion that functions through delayed privatization. Its spatial variation, similar to that of the TRI, is displayed in [Figure 8](#).

PSD vote share, an endogenous, but treatment-relevant variable, is utilized as a proxy for political capture given that, as the institutional successor to the communist apparatus, it distinctly relied on rural local networks and state patronage for electoral control (Mares and Young, [2019](#)).²⁰ It is measured as the percentage of valid votes cast for the Social Democratic Party in the first round of Romanian legislative elections in order to isolate party-specific loyalty from individual-candidate-specific factors and protest votes. Legislative election results at the county level are available from the Permanent Election Authority ([2026](#)) for the election years 1996, 2000, 2004, 2008, 2012, and 2016. The 2012 election is completely excluded from the variable sample due to the party running under a different coalition label that year, which would introduce a mechanical discontinuity in the series. Because elections occur every four years, the OLS specification uses observations only for those years during which elections were held. For the split-sample DiD and synthetic DiD, the vote share functions as a moderator rather than an outcome. Each county’s mean PSD share across all available elections is computed, and counties above the full sample median of this time-invariant mean are classified as High-PSD, those below as Low-PSD ([Figure 9](#)). This partition is designed to detect structural political capture, which is the stable, decade-spanning dominance of clientelist network, rather than electoral volatility in any single cycle. In the OLS baseline, PSD vote share also appears as an outcome variable to establish the geographic baseline association between Plains status and political capture.

²⁰In a few cases, however, using PSD vote share as a proxy for clientelism may be limited by demographic factors, such as the distribution of ethnic Hungarian minorities, who are concentrated in Transylvanian Mountains counties. Because this demographic votes often for the Democratic Alliance of Hungarians in Romania (UDMR), low PSD vote shares in these specific counties could partially reflect ethnic electoral sorting rather than an absence of support for the PSD.

The primary outcome variable for the asset mismatch channel is tractor density, measured as the number of tractors per 100 hectares of arable land at the county level.²¹ The normalization by arable land rather than total county area or population is deliberate. Given that the asset mismatch mechanism predicts a deficit in agricultural capital investment specifically, arable land, hence, is the relevant denominator for measuring whether the capital stock is adequate for the farming activity the terrain supports. The cleaned sample excludes two extreme outliers of Ilfov and Bucharest, which, in particular, has its tractor density more than ten times larger than the rest of the counties. Its spatial variation and that of some other outcome variables can be found in the [Appendix B](#) of this paper.

The remaining four outcome variables capture the downstream demographic and political consequences of the asset mismatch and political capture mechanisms. Rural business formation measures rural firm registrations per 1,000 permanent countryside inhabitants, where both enterprise and inhabitant data were sourced from INS (2026) registry. The countryside restriction — firm registrations in localities classified as rural by Romanian administrative law — is key. Urban business formation within Plains counties would conflate the agricultural asset mismatch mechanism with urban agglomeration effects unrelated to collectivization history. To explicitly capture the political mechanism of the institutional trap, the fiscal dependency ratio measures the percentage of central state subsidies over total local revenues. High levels of fiscal reliance serve as a proxy for a county's vulnerability to the clientelist resource allocation of the PSD machine.

As far as demographics are concerned, rural natural increase measures the balance of births and deaths per 1,000 permanent countryside inhabitants. The rural restriction here follows a similar logic as for businesses. The demographic mechanism should operate through immobility in agricultural communities specifically, and total natural increase would dilute this signal with urban dynamics. Finally, external migration rate measures net international migrants per 1,000 permanent inhabitants calculated using the following formula:

$$\text{Net Migration Rate}_{ct} = \frac{\text{Arrivals}_{ct} - \text{Departures}_{ct}}{\text{Population}_{ct}} \times 1,000$$

Negative values of this measure indicate net emigration and vice versa. The only limitation of this metric is that it is measured at the total county level instead of the rural population as Romanian administrative records do not disaggregate international migration flows by rural-urban origin, which is a limitation shared across the economics literature on Romania (Sandu, 2005). Nevertheless, the spatial variation of the total measure, especially its evolution over-time, seems as a relatively reliable indicator of the immobility trap and the liquidity constraint following shocks, such as visa-free regime announcement and EU accession that coincide with larger international emigration in the Mountains and less so in the Plains ([Figure 10](#)).

²¹Tractor stock and arable land area data are sourced from INS (2026). The per-hectare normalization is preferred over a per-capita measure because the mechanism operates through agricultural capital intensity rather than population density. A county with few tractors relative to its arable area is one that plausibly has not fully mechanized its agriculture, regardless of how many people live there.

All specifications, except the synthetic DiD one, include four time-varying socioeconomic controls selected to account for pre-existing county-level development differences that could confound the Plains-Mountains comparison. Modernized roads ratio measures the percentage of modernized and light asphalt roads in the total county road network, also sourced from the INS (2026), and proxies for the quality of transport infrastructure. A county with better roads might show different business formation and fiscal dependency independently of its agricultural history. High school graduates per 1,000 permanent inhabitants ratio controls for human capital differences that could independently affect both migration and business formation outcomes. Hospital beds index measures the number of beds in hospitals per 1,000 permanent inhabitants and indicates public service provision quality. It is a control for the possibility that fiscal dependency reflects genuine service delivery differences rather than clientelist capture. Non-political social aid index is measured by weighting aggregate spending on unemployment benefits by the permanent resident population in a county. It is specifically chosen over broader social transfer measures because it may isolate automatic, eligibility-based transfers rather than discretionary political allocations. The latter would be endogenous to the political capture mechanism the paper is testing.

Table 3 presents summary statistics for the cleaned sample, while the pre-processed descriptive statistics can be found in Appendix A. Panel A of the table reports on the full sample, whereas Panel B provides a Plains-Mountains comparison with t-tests for mean differences. Several patterns in Panel B are directly relevant to the identification strategy and warrant attention before the empirics. First, the tractor deficit is large and precisely estimated: Plains counties average 1.71 tractors per 100 arable hectares against 3.86 in the Mountains, a difference significant at the 1% level, which is a raw descriptive gap that the regression analysis may then attribute causally to the asset mismatch mechanism. Second, the privatization gap is similarly significant at 1% no matter what time frame is being taken and that is consistent with the predicted effect of the governance vacuum over collective infrastructure on the pace of land market development. Third, rural business formation seems to run in the opposite direction with Plains having more firms per 1,000 permanent inhabitants than the Mountains, significant at 5%, suggesting that the unconditional Plains effect is positive and that suppression in the Plains might only be recoverable conditional on PSD capture intensity.

Net migration, although confirmed that it is higher in the Mountains, thus, enforcing the immobility trap theory in the Plains, does not indicate a significant difference after a t-test, which validates the guess about ambiguous effects on migration. Insignificant difference for rural natural increase and fiscal dependency motivates the need to delve into the political capture mechanism in the identification strategy, too. At the same time, PSD vote share being significantly higher in the Plains counties helps to establish the geographic baseline association between terrain and political capture. As for controls, the most notable differences are in high school graduation rates and the social aid index, which are significantly higher in the Mountains, probably reflecting the historically stronger non-political welfare infrastructure in Transylvanian counties. Hence, inserting them into regression specifications should aid in preventing omitted variable bias, especially in the baseline OLS specification. Nevertheless, while road quality and hospital beds density show no significant unconditional difference, including them

should instead help absorb residual variance and further ensure that causal identification with subsequent panel models relies on parallel trends instead of initial level equivalence.

V. Identification Strategy

The empirical analysis proceeds in three stages to identify the legacy of Romania’s agricultural decollectivization. First, baseline OLS panel regressions establish structural correlations between the geographic asset mismatch, privatization rates, and other outcomes. To address unobserved time-invariant heterogeneity, the analysis then employs a dynamic event study leveraging the exogenous shock of the 2007 EU accession. Finally, to isolate the mechanism of political capture, the strategy applies split-sample and synthetic difference-in-differences estimators (Arkhangelsky et al., 2021). Across the models, causal identification relies fundamentally on the strict exogeneity of the geographic treatment (terrain ruggedness), which is acting as a permanent natural endowment orthogonal to 1990s transition policies, as well as the Stable Unit Treatment Value Assumption (SUTVA), presuming no widespread spillovers from the Plains to the Mountains control group.

A. Ordinary Least Squares

The analysis begins by estimating the persistence of the agricultural asset mismatch across a standard cleaned panel of 40 counties. To first establish that geographic fundamentals could have inherently influenced the speed of economic transition, the annual privatization rate (Pr_{ct}) is regressed directly on the geographic treatment, conditioning on initial baseline disparities:

$$Pr_{ct} = \alpha_0 + \rho \cdot Pl_c + \gamma_1 \cdot \mathbf{X}_{c,1996} + \delta_t + \varepsilon_{ct} \quad (1)$$

Next, the primary structural OLS specification evaluates how this geographic and institutional mismatch translates into downstream rural and political outcomes:

$$Y_{ct} = \alpha_1 + \beta_1 \cdot Pl_c + \beta_2 \cdot Pr_{ct} + \gamma_2 \cdot \mathbf{X}_{c,1996} + \delta_t + \varepsilon_{ct} \quad (2)$$

where Y_{ct} represents the outcome of interest, such as tractor density, migration, and others in county c at time t . The variable Pl_c is a time-invariant binary treatment indicator taking the value of 1 if the county’s mean TRI falls below the sample median, capturing the flatlands susceptible to the capital-land mismatch. Pr_{ct} tracks the rate of agricultural land privatization. To absorb aggregate macroeconomic shocks affecting all counties, such as the 1999 recession or 2007 EU accession, both models include year fixed effects (δ_t).

A critical identification choice in both specifications is the inclusion of $\mathbf{X}_{c,1996}$, a vector of socioeconomic covariates, such as modernized roads, high school graduation rates, and others. They are strictly fixed at their 1996 baseline levels. Including time-varying controls in regressions evaluating a long-term transition process risks introducing post-treatment bias (“bad controls”). Because infrastructure, education, and welfare may naturally evolve as consequences of a region’s economic trajectory and privatization speed, controlling for their

contemporaneous values would artificially absorb the exact mechanisms this paper measures. Fixing these controls helps isolate the geographic penalty from historical baseline differences, such as the stronger institutional infrastructure in Transylvanian Mountains counties, without over-controlling for the downstream effects of the policy shock.²²

Finally, to account for serial correlation within regions over time, standard errors are clustered at the county level. The main text reports the full specifications, while unadjusted baseline models (without controls) are provided in [Appendix C](#).

B. Baseline Difference-in-Differences and Dynamic Event Study

To move from structural correlations to causal identification, the analysis leverages the exogenous macroeconomic shock of Romania's 2007 EU accession. By gradually lifting mobility barriers starting in 2002 with the new visa-free regime and by abruptly removing trade restrictions in 2007, EU integration helped to reveal the latent vulnerability of the Plains' agricultural mismatch. It can be well captured with the Average Treatment Effect on the Treated (ATT) across the entire post-accession period that will require a standard Two-Way Fixed Effects (TWFE) model to be estimated:

$$Y_{ct} = \beta \cdot (Pl_c \times Post_t) + \gamma \cdot \mathbf{X}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \quad (3)$$

where Y_{ct} again denotes the outcome for county c at time t , $Post_t$ is an indicator taking the value of 1 for the years 2007 onward, and β captures the pooled divergence. To observe the precise temporal evolution of this divergence and visually verify the parallel trends assumption, this framework is expanded into a dynamic event study:

$$Y_{ct} = \sum_{k \neq 2006} \beta_k \cdot (Pl_c \times \mathbf{1}_{t=k}) + \gamma \cdot \mathbf{X}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \quad (4)$$

where the dynamic parameters of interest, β_k , estimate the annual gap between the Plains and Mountains relative to the pre-treatment base year of 2006.

A key methodological change here is the inclusion of county fixed effects (α_c). In the baseline OLS model, county fixed effects had to be omitted because they would perfectly absorb the time-invariant geographic treatment (Pl_c). In this panel framework, however, the treatment effect is isolated through the interaction term. This allows α_c to rigorously control for all unobserved time-invariant heterogeneity, such as historical culture and permanent soil quality, without absorbing the divergence of interest. Year fixed effects (δ_t) are retained to absorb

²²Here, the standard panel utilizes N=760 annual county-years because privatization rates contain data only until 2014, while the PSD vote share regressions rely on N=160 county-election years that naturally exclude the 2016 election. The rural natural increase model operates with N=759 due to a missing historical observation for Neamț in 1997, and rural business formation utilizes N=720 as business registry data begin post-1996. For the same reason, hospital bed density cannot be included in the set of control variables in the OLS setting.

aggregate national shocks, such as the 2008 global financial crisis.²³

This fixed effects structure also necessitates a shift in the control vector ($\mathbf{X}_{ct}$). Given that the static 1996 controls used in the OLS regressions are perfectly collinear with α_c , time-varying controls are introduced to account for localized developmental trajectories. To address the econometric risk that these time-varying covariates might act as endogenous “bad controls” following the EU shock, all models are estimated both with and without $\mathbf{X}_{ct}$ (Appendix C) to prove the treatment effect is not an artifact of post-treatment conditioning.

As mentioned before, causal interpretation here relies on the parallel trends assumption. The event study (Equation 4) directly tests this by visualizing the pre-trends. Standard errors are again clustered at the county level to correct for arbitrary serial correlation.

C. Split-Sample Difference-in-Differences

While the baseline models isolate the 2007 EU accession as the important inflection point of regional divergence, they probably cannot inherently distinguish whether this stagnation was driven purely by the economic friction of the agricultural asset mismatch or whether it was actively exacerbated by political capture. To isolate the mechanism of local political clientelism, the empirical strategy employs a split-sample difference-in-differences design.

By partitioning the treated Plains counties based on their vulnerability to the political machine, which is proxied by their mean PSD vote shares, this design tests whether the intensity of the post-2007 divergence scales directly with the strength of local clientelist networks. To do this, the pooled TWFE specification (as in Equation 3) is estimated separately for the High-PSD and Low-PSD subsamples to capture the distinct ATTs for each political cohort. Additionally, the dynamic event study framework (Equation 4) is applied to each subsample to visually confirm the parallel trends assumption within each specific split.

The core intuition of this split-sample approach relies on maintaining a consistent control group, while varying the treatment intensity. Both the High-PSD and Low-PSD treated counties are benchmarked against the exact same 21 Mountains counties, which historically feature robust non-PSD competition and alternative economic safety nets.²⁴

Finally, recent econometric advancements highlight that the standard TWFE estimator can be strongly biased in the presence of staggered treatment timing and heterogeneous treatment effects across cohorts (de Chaisemartin & D’Haultfœuille, 2023). Fortunately, because the identifying shock of this paper (EU accession) occurs simultaneously for all treated units in the single year of 2007, this design inherently avoids negative weighting issues associated with staggered rollouts. Consequently, valid causal identification in this standard 2×2 frame-

²³The panel has N=960 annual county-years. Full specifications incorporating the hospital beds index or rural business formation, however, utilize slightly reduced panels (N=920 or 919) due to data availability constraints.

²⁴The sample splitting is based on the median PSD vote share among all 42 counties. Thus, High-PSD sample pairs 13 High-PSD Plains with the 21 Mountains (N=816 county-years; N=782 when including time-varying controls and rural business formation). The Low-PSD sample pairs the remaining 6 Low-PSD Plains with the same 21 Mountains (N=648 county-years; N=621 with full controls). Variables with missing historical data, such as rural natural increase, result in marginally lower observation counts.

work requires only the parallel trends assumption, the plausibility of which will be checked with the help of the visualization of pre-trends. Standard errors remain clustered at the county level.

D. Synthetic Difference-in-Differences

Even though the split-sample TWFE design should effectively help to capture the political mechanism, slicing the dataset into smaller political sub-cohorts introduces distinct econometric challenges. Specifically, the High-PSD sample relies on comparing just 13 treated Plains counties against the 21 Mountains controls. In such small-N environments, standard TWFE estimators can be inefficient and highly sensitive to the parallel trends assumption, as equally weighting all 21 Mountains counties may incorporate noisy or poorly-matched counterfactuals. Furthermore, heavily parameterized models with time-varying controls sometimes risk small-sample overfitting.

To resolve these limitations and rigorously validate the clientelism mechanism, the final stage of the empirical strategy applies the synthetic difference-in-differences estimator, developed by Arkhangelsky et al. (2021), primarily to the High-PSD sample, alongside a parallel estimation on the Low-PSD sample to serve as a falsification test to check robustness. SDiD improves upon both standard DiD and traditional synthetic controls methods by simultaneously estimating unit weights and time weights to construct an optimal, uncaptured counterfactual. The estimator is defined as:

$$(\hat{\tau}^{SDiD}, \hat{\mu}, \hat{\alpha}, \hat{\delta}) = \arg \min_{\tau, \mu, \alpha, \delta} \sum_{c,t} (Y_{ct} - \mu - \alpha_c - \delta_t - \tau W_{ct})^2 \hat{\omega}_c \hat{\lambda}_t \quad (5)$$

where W_{ct} is the binary treatment indicator taking the value of 1 for the treated Plains sub-cohort post-2007, $\hat{\tau}^{SDiD}$ represents the causal treatment effect of EU accession on that specific cohort, and μ captures the grand mean (overall intercept). By estimating separately for the Low-PSD sample, the framework mathematically verifies whether the divergence changes in the absence of the political machine.

The strength of the SDiD estimator lies in its dual-weighting mechanism. First, it calculates unit weights ($\hat{\omega}_c$), which are assigned exclusively to the donor pool (the 21 Mountains counties).²⁵ Instead of treating all Mountains counties equally, the algorithm mathematically selects and reweights a specific combination of counties to synthesize a “perfect” control group whose pre-2007 trajectory precisely mimics that of the High-PSD Plains. Second, it calculates time weights ($\hat{\lambda}_t$), which are applied to the pre-treatment periods to more heavily weight the specific pre-shock years that are most predictive of post-shock outcomes, effectively stabilizing the pre-trend matching.

²⁵The High-PSD estimation uses 13 treated Plains and 21 Mountains donors over 24 years (N=816). the Low-PSD falsification uses 6 treated Plains and the same donors (N=648). Business formation N is lower due to missing early-panel data. To satisfy SDiD’s balanced panel requirement, a single missing 1997 Neamt observation was linearly interpolated.

Consequently, this dual-weighting optimization renders the inclusion of the socioeconomic control vector ($\mathbf{X}_{ct}$) unnecessary. Given that the synthesized control group is constructed to perfectly match the treated group’s pre-treatment structural evolution, the weights naturally absorb the confounding localized trajectories that the covariates previously attempted to proxy. By dropping the controls, the SDiD framework helps to eliminate the risk of post-treatment bias and small-sample overfitting.

Finally, unlike standard TWFE models that rely on clustered standard errors, statistical inference in this SDiD framework is conducted using the Jackknife method. By iteratively dropping single units and re-estimating the model, the Jackknife particularly accounts for the variance introduced by the estimation of the weights themselves, providing more conservative and reliable standard errors in a restricted 34-county panel.

VI. Results

A. Geographic Baseline

Establishment of the structural baseline relies on the OLS estimates of the geographic penalty, which are found in [Table 4](#). As previously mentioned, this and other models estimated with no controls can be found in [Appendix C](#). Coefficients for the binary Plains dummy represent average level differences versus the Mountains, interpreted in native units, which are percentage points (privatization, PSD vote, fiscal dependency), units per 100 arable hectares (tractors), and per 1,000 permanent inhabitants (businesses, migration, natural increase).

The table hints at a considerable institutional and physical capital penalty in the Plains. The geographic dummy predicts a privatization rate 3.38 percentage points lower than the Mountains ($p < 0.01$), confirming a pre-existing spatial disparity. This relatively stalled reform coincides with a substantial structural deficit of -1.78 tractors per 100 arable hectares that translates into a 68% reduction relative to the sample mean. Notably, the privatization rate itself does not predict tractor density, which suggests that the capital deficit is a direct geographic consequence of the asset mismatch (the hold-up problem), not merely a symptom of slow administrative reform.

Conversely, enterprise creation reveals a counterintuitive mechanism. While geography alone does not significantly suppress rural business formation, the privatization rate, surprisingly, does. Although the coefficient seems small (-0.022), the fact that the average privatization rate varies across the Plains and the Mountains by up to five percentage points — even more if taking individual counties — makes this a meaningful magnitude. Because the Plains experienced lower privatization, this negative coefficient presents an empirical puzzle, which indicates that the flatlands might have actually possessed a latent structural advantage for enterprise creation. This suggests a complex trade-off. Even though physical capital destruction was driven directly by the geographic asset mismatch, the suppression of rural businesses cannot be explained by delayed reform, leaving an anomaly that geography seems to have failed to resolve.

Finally, the estimates establish the spatial foundations of both political capture and demographic immobility. The Plains dummy predicts a 7.06 percentage point increase in PSD vote share, a relative 20% increase over the mean. Simultaneously, the positive external net migration coefficient (0.354) indicates comparatively lower emigration or even higher immigration to the Plains. This empirically supports the liquidity constraint hypothesis, where asset-rich, but cash-poor households lacked the liquid means for international exit.

However, core indicators, such as rural natural increase and fiscal dependency, yield insignificant geographic baseline effects. Geography plausibly laid the trap and initialized the lock in immobility, but realizing the ultimate demographic and fiscal stagnation might have required exogenous 2007 EU accession shock and further enforcement of political capture.

B. Asset Mismatch Channel

Having established the geographic baseline, it is useful to now turn to the dynamic macroeconomic shock initialized by Romania joining the European Union. [Figure 11](#) and [Table 5](#) help evaluate whether integration into the European single market resolved Plains' asset mismatch or accelerated its collapse.

[Figure 11](#) (left column) visualizes the dynamic event study trajectories across three distinct institutional phases: restricted mobility (1996–2001), the visa-free intermediary regime (2002–2006), and post-EU accession (2007 onwards). For modern developmental indicators, such as business formation, natural increase, and, to an extent, fiscal dependency, the Plains and Mountains exhibit parallel pre-trends across both pre-shock phases. Tractor density exhibits a slight downward pre-trend from 1996 to 2001, reflecting the steady depreciation of the inherited socialist fleet without replacement. However, this trajectory stabilizes onto the zero-line during the 2002–2006 intermediary period. This confirms that on the eve of EU accession, parallel trends generally held across all core outcomes.

Once formal accession occurred, the trade shock acted as a significant catalyst for agricultural divergence. Column (1) of [Table 5](#) confirms that exposing the Plains to open European competition influenced a persistent fall in physical capital, dropping tractor density by an additional 1.56 units per 100 hectares ($p < 0.01$). The exogenous market shock seems to have exposed the vulnerability of the fragmented agricultural system, aggravating the decay of the region's capital stock. Yet, interestingly, the aggregate pooled DiD reveals a statistically significant increase in rural business formation (0.300). This complements the empirical paradox: if the core agrarian foundation of the Plains was collapsing, why was aggregate enterprise creation rising?

The dynamic post-treatment graphs for demographics and local institutions deepen this puzzle. While their pooled estimators lack statistical significance, their visual trajectories reveal distinct structural patterns shaped by broader macroeconomic headwinds. Rural natural increase experiences a clear downward post-shock trend, indicating a developing demographic decay. Concurrently, net migration trends upward, that suggests comparatively less emigration or more return migration to the Plains although the wide confidence intervals reflect high spatial variance. It should be taken into account, however, that the variable's response to the shock

seems to have been delayed likely by the 2008 global financial crisis as well as the 2012 European debt crisis, which were associated with contracted labor demand in primary European destinations for Romanians, such as Italy and Spain. Consequently, these events may have temporarily trapped potential migrants who might otherwise have utilized their newly acquired mobility rights and lower exit costs.

Finally, fiscal dependency traces a distinct U-shaped curve. The initially high dependency falls steadily during the early 2000s as Romania implemented relatively strict macroeconomic stabilization and austerity policies required to qualify for EU membership. Then it remains flat immediately following accession, which is expected as the 2008 crisis suppressed state revenues uniformly, but begins trending upward after 2012. As the aforementioned Eurozone sovereign debt crisis took place, it probably strained external remittance channels, and as EU integration expanded the central resources available for local patronage, reliance on state transfers in the structurally weak Plains may have swelled once again.

Ultimately, these noisy aggregate demographics, the fiscal U-curve, and the paradoxical rise in businesses indicate that evaluating the Plains as a monolithic geographic block may not be analytically complete. The aggregate treatment effect plausibly hides important institutional heterogeneity. To understand how the physical capital collapse might have been actively instrumentalized into an immobile, dependent Fiscal Zombie equilibrium, the analysis moves beyond geography and isolates the local political apparatus.

C. Political Capture Channel

[Figure 11](#) (right column) and [Table 6](#) attempt to resolve the aforementioned puzzle by partitioning the Plains into High-PSD and Low-PSD sub-cohorts, effectively revealing that the aggregate treatment effect hides two different institutional trajectories.

The split-sample event studies demonstrate that the destruction of the agrarian base was mostly geographic, but the subsequent recovery was rather political. Looking at tractor density in [Figure 11](#), the trajectories of both cohorts are virtually identical. The slight pre-2002 aggregate parallel trend violation is shown to be generally driven by the High-PSD cohort, which possessed a larger initial stock of aging machinery. Post-2007, both cohorts experience a nearly identical, statistically significant collapse in their pooled tractor density coefficients.

However, a divergence emerges in enterprise creation. While both cohorts exhibit parallel pre-trends, the pooled estimator for post-shock business formation increases relatively significantly in the Low-PSD cohort. Conversely, the High-PSD trajectory is more muted, remaining lower and frequently touching the zero bound. This helps to further dissect the aggregate puzzle. When the agricultural base collapsed, politically competitive counties successfully transitioned toward, for example, non-farm commercial enterprises. In captured counties, the political machine may have influenced the stifling of this commercial exit strategy, trying to preserve a monopoly on local economic opportunity.

By blocking enterprise, the political apparatus likely contributed to honing a demographic trap. In the Low-PSD cohort, rural natural increase and net migration remain quite flat and stable post-shock. Notably, Low-PSD natural increase violates pre-trends by converging diagonally

before 2007, but stabilizes completely along the zero line post-accession. In contrast, the High-PSD pooled coefficient for natural increase captures a considerable and statistically significant demographic decay. At the same time, High-PSD net migration experiences an upward jump in magnitude, almost mirroring the aggregate event study, though its pooled coefficient remains statistically insignificant due to wide spatial variance.²⁶

The objective of this demographic trap is visible in the fiscal dependency trajectories. The Low-PSD cohort experiences relative fiscal stability, hovering around the zero line from 1997 through 2018. The delayed aggregate U-curve identified in the previous subsection was possibly fueled by the High-PSD cohort. As the captured population probably became increasingly locked-in, High-PSD dependency started to increase after 2012. While the overall pooled coefficient remains statistically insignificant, diluted by the flatline of the 2008 crisis, the dynamic point estimates become significant by 2018.²⁷

Hence, the split-sample visually dissects the anatomy of the Fiscal Zombie equilibrium. Geography seems to have destroyed the agricultural base, but it was the PSD apparatus that could have taken advantage of this collapse, thus, suppressing enterprise to produce a captive, immobile, and state-dependent electorate. However, because standard TWFE estimators struggle to achieve strict statistical significance across multiple pooled variables in a restricted, small-N split-sample, the final stage of analysis applies the SDiD estimator to High-PSD Plains in order to definitively eliminate pre-trend noise and rigorously quantify the causal magnitude of this political trap.

D. Quantifying the Fiscal Zombie

Figure 12 plots the observed trajectories against their optimized synthetic counterfactuals not only for the main High-PSD specification, but also for Low-PSD as a falsification test. Formal estimates for the High-PSD cohort are presented in Table 7. The coefficients for the falsification test can be found in Appendix D.

The SDiD framework confirms that the decay of physical capital was, in effect, a universal geographic penalty in the Plains, but the suppression of enterprise probably was a deliberate political choice. Post-2007, the High-PSD tractor trajectory flatlines, resulting in a significant relative penalty (-0.994 , $p < 0.01$), although smaller in magnitude than previous models as the synthetic Plains counterfactual begins re-mechanizing. While the Low-PSD falsification cohort also stagnates, it stabilizes at a slightly higher threshold.

Political capture mechanism becomes more evident when observing rural business formation. The Low-PSD falsification trajectory shows that the Plains may have possessed a considerable latent commercial advantage, surging well above its synthetic counterfactual post-shock.

²⁶While this localized upward migration trend strongly supports the liquidity constraint hypothesis, it may also be partially inflated by the post-2007 influx of Moldovan citizens acquiring Romanian passports and settling in eastern border counties. Nevertheless, the structural immobility of the native population remains quite clear.

²⁷Methodologically, the exceptionally high aggregate R^2 for fiscal dependency across all specifications, including the baseline OLS, presumably reflects the deeply persistent, structural nature of local state revenues. The strictly bounded adjusted within- R^2 confirms that this high aggregate fit successfully absorbs baseline heterogeneity rather than indicating multicollinearity among the covariates.

Conversely, the High-PSD cohort's business formation is lower and quite similar to that of the synthetic control. While it manages a statistically significant positive effect against its synthetic baseline (0.213), its slope and magnitude are rather suppressed relative to the region's non-farm potential. The local political apparatus might have allowed just enough enterprise to function, but essentially tried to restrain structural commercialization to maintain its monopoly on economic leverage.

By influencing this commercial exit, the machine seems to have finalized a demographic trap. Even though the Low-PSD cohort tracks its synthetic demographic counterfactual, the High-PSD cohort experiences a decrease. Rural natural increase breaks from the synthetic baseline post-accession, falling into a worsening decay (-1.270) equivalent to approximately 30% of its pre-treatment mean. Simultaneously, net migration witnesses a visible post-2010 upward increase against its synthetic control, indicating a reduction in out-migration or increase in immigration. Together, these two dynamics aid in describing a demographic reality generally defined by captured and immobile population that is rapidly aging and no longer reproducing.

One of the key aims of this trapped, dying demographic base is realized in the fiscal trajectories. While the Low-PSD cohort becomes relatively more fiscally independent than its synthetic counterfactual, the High-PSD cohort seems to diverge upward from its synthetic baseline after 2014 the same way it did in the previous subsection. With statistical noise mathematically eliminated, the SDiD estimator suggests a considerable and statistically significant increase in local fiscal dependency by 1.83 percentage point. Against a pre-treatment mean of 13.20%, this, thus, represents a relative structural shift toward more state reliance by an increase of about 14%.

VII. Robustness Checks

In order to verify that the final equilibrium is not an artifact of sample selection or model specification, it is important to establish essential robustness checks in addition to the falsification test discussed previously. Hence, this section subjects the core empirical mechanisms to four sensitivity tests. The detailed results of these tests can be found in [Appendix D](#).

A. Extreme TRI Split

First, a common threat to spatial DiD designs is the potentially arbitrary nature of geographic boundaries. The binary Plains against Mountains classification risks mischaracterizing counties situated in transitional, hilly regions. If these middle-tier counties are misclassified, they could artificially drive or suppress the aggregate geographic penalty observed in the estimates.

To resolve this boundary ambiguity, the TRI is utilized to execute an extreme subsample analysis. The middle 50% of the aggregate sample is strictly dropped, isolating and re-estimating the baseline DiD model exclusively on the flattest 25% (ten extreme Plains counties) against the most rugged 25% (ten extreme Mountains counties).

The results in [Appendix D](#) confirm the geographic origins of the asset mismatch. Eliminating transitional geographies magnifies the baseline physical capital penalty. The extreme Plains exhibit a post-2007 tractor deficit of -2.37 ($p < 0.01$) with full controls, a substantial intensification compared to the baseline aggregate penalty of -1.56. This suggests that the agricultural collapse is plausibly a structural geographic phenomenon rather than a fuzzy boundary artifact. Interestingly, rural business formation in these flattest counties also increases significantly (0.501). Given the previous finding that enterprise creation was influenced to a great extent by politically competitive (Low-PSD) counties, this amplified coefficient indicates that the least rugged Plains actually might possess the largest latent potential for non-farm commercialization. Simultaneously, demographic and fiscal outcomes remain statistically indistinguishable from zero across this extreme spatial divide. This supports one of the premises of the paper that, while extreme terrain influenced the initial capital mismatch and the destruction of heavy agriculture, geography alone does not explain the subsequent demographic trap.

B. Extreme PSD Split

Even though the main split-sample analysis helped to establish the presence of political capture, categorizing all Plains counties as simply High or Low PSD risks diluting the real magnitude of the mechanism. Counties hovering near the median political threshold might blur the institutional divergence. If the Fiscal Zombie theory holds, the suppression of enterprise and the resulting demographic collapse should act as a dose-response mechanism: scaling in severity alongside the intensity of clientelist capture.

To test this, the extremes of the political spectrum within the Plains are isolated. The middle 50% of the Plains counties are removed, running the split-sample model strictly on the top 25% (five severely captured counties) and the bottom 25% (five uncaptured counties) against the standard Mountains control group.

The extreme dose-response estimates in [Appendix D](#) lend support to the anatomy of the clientelist trap. The decay of physical capital remains universal, yielding slightly more severe or unchanged, but still statistically significant tractor deficits in both heavily captured (-1.563) and uncaptured (-1.500) extremes. However, the economic recovery is rather bifurcated. Rural business formation increases significantly (0.520) in the uncaptured Plains, but is indistinguishable from zero in the captured Plains, confirming the original findings. Consequently, with less heavy agriculture and independent enterprise, the extreme PSD Plains experience an intensified collapse in rural natural increase, which is even greater than in the main findings (-2.322 against -1.801), whereas the uncaptured Plains remain more stable. By isolating the extreme tails of political control, stable or even more severe coefficients help to show that clientelist networks might have contributed to gatekeeping EU-era economic adaptation to sustain a captive demographic base.

C. Truncated Timeline

Nevertheless, the fact that the first decade of Romania's post-communist transition was quite volatile needs to be taken into account, too. The 1990s were characterized by severe macroeconomic instability, hyperinflation, and chaotic deindustrialization (Gallagher, 2005; Pop-

Eleches, 2007). Thus, the observed divergence might simply capture a delayed hangover from this turbulent decade rather than a true structural response to the 2007 EU accession.

In order to separate the EU integration era from early-transition noise, the dataset is permanently truncated by dropping all observations prior to 2000. The baseline DiD model is then re-estimated exclusively on the restricted 2000–2019 window.

As reported in [Appendix D](#), dropping the unstable 1990s leaves the core mechanisms statistically undisturbed in general. The tractor density penalty remains highly significant and changed only marginally (-1.460 , $p < 0.01$), confirming that the agricultural divergence was affected by open market exposure instead of 1990s events. Furthermore, the structural adaptation toward rural enterprise remains present (0.273), supporting the theory that independent entrepreneurship was likely a recent adaptation incentivized by EU accession. Core demographic and fiscal parameters maintain their baseline insignificance. Overall, the restricted timeline confirms that the capital collapse and subsequent institutional divergence are relatively reliable functions of the post-accession era.

D. Leave-One-Out Analysis

Finally, in a geographically restricted panel ($N=40$ counties), baseline estimates risk being artificially driven by a single high-leverage outlier. This is a particular concern for the external migration parameter given that it is possible that a single eastern border county could be skewing the aggregate migration variance due to the localized post-2007 influx of Moldovan citizens acquiring Romanian passports.

To test for outlier sensitivity, one county at a time is fully dropped from the panel and the baseline model is iteratively re-estimated, generating 40 distinct point estimates for each dependent variable, which are visualized in [Figures D1–D5](#) of [Appendix D](#).

The Leave-One-Out (LOO) plots demonstrate that the baseline findings are stable and presumably systemic. For physical capital decay, the tractor density point estimate hovers reliably around -1.56 , with the 95% confidence intervals excluding the zero threshold regardless of which county is taken out of the sample. The positive rise in aggregate business formation is equally stable. Rural natural increase and fiscal dependency parameters remain indistinguishable from zero no matter which county is excluded.

Furthermore, the LOO analysis helps to address the migration anomaly. Iteratively dropping individual eastern border counties, such as Vaslui, Iași, or Galați, does not alter the wide confidence intervals or the statistical insignificance of the net migration rate. This suggests that the high variance in migration is not merely a statistical artifact of a single border anomaly. The agricultural stagnation and subsequent economic bifurcation in the Plains seem to be robust macro-level trends, likely independent of any single local administration.

VIII. Conclusion

Romania's transition from 1950s forced collectivization to European Union integration left behind a noticeably unequal and structurally fractured rural landscape. This thesis, in particular, investigated the long-term consequences of the 1991 land restitution, identifying how geographic constraints and local political capture interacted following the exogenous macroeconomic shock of the 2007 EU accession. By utilizing baseline and synthetic DiD frameworks, the analysis tackled a particular puzzle: why the agricultural Plains fell into demographic decay and fiscal dependency, while the rugged mountainous regions structurally adapted relatively better.

The empirical findings show that the collapse of rural physical capital was a sizable geographic penalty. The egalitarian fragmentation of land in 1991 may have influenced the creation of an asset mismatch in the Plains, significantly reducing the productive value of indivisible socialist agricultural machinery and resulting in an approximately 68% baseline collapse in regional tractor density relative to the mean. However, the subsequent economic trajectory of these regions seems to have been rather determined by local political institutions. Even though politically competitive Plains relatively successfully transitioned toward non-farm rural enterprises, clientelist networks in captured counties were associated with considerably weaker commercial adaptation, with business formation approximately half that of their less-captured Plains counterparts, based on the synthetic DiD estimates. Consistent with a Fiscal Zombie equilibrium, counties with deeper political capture showed patterns of gatekept EU structural funds and suppressed private enterprise, as well as, on average, a 14% relative increase in local state reliance. The results conform to an immobile, aging population becoming increasingly dependent on state transfers, supporting the hypothesis of a captive electorate. Consequently, this paper contributes to the broader political economy literature by showing that, while geography plausibly influences the anatomy of an initial economic shock, political capture may affect the possibility of long-term structural survival.

Despite the robust findings, the analysis faces limitations that offer fertile ground for future research. Primarily, the demographic mechanism, specifically the external migration parameter, apart from being a total and not a rural variable, is potentially affected by spatial noise due to the post-2007 influx of Moldovan citizens obtaining Romanian passports and settling in eastern border counties. Future empirical work could develop granular filtering mechanisms to separate native rural out-migration from this cross-border administrative artifact. A further limitation concerns the use of PSD vote share as a proxy for political capture, which remains potentially endogenous to the same economic stagnation it is meant to explain. Thus, future research could address this by instrumenting PSD dominance with predetermined measures of communist organizational density, such as county-level party membership rates or *Securitate* officer presence in the 1950s–60s, which could predict clientelist network persistence through institutional continuity, while remaining orthogonal to post-accession outcomes. Moreover, exploring the broader regional context through comparative studies with other post-communist states, such as the Republic of Moldova or Bulgaria, could help determine whether the clientelist demographic trap is a uniquely Romanian institutional legacy or a general feature of delayed agrarian transitions.

Finally, to fully resolve the institutional puzzle of Romanian agriculture, further research could examine the historical prequel to this study, which is the violent collectivization process itself. As data and historical context provided in [Section II](#) showed, there is a promising avenue for future doctoral research that could involve leveraging newly accessible archival data on peasant arrests, resistance, and forced collectivization quotas. Quantifying the initial destruction of human and economic capital during the 1950s would aid in connecting the historical dots, mapping the complete trajectory of the Romanian village from its initial communist takeover to its modern post-communist bifurcation.

References

- Acemoglu, D., Johnson, S. & Robinson, J. A.** (2001) The Colonial Origins of Comparative Development: An Empirical Investigation. Cambridge, Mass: *National Bureau of Economic Research*.
- Adamopoulos, T. & Restuccia, D.** (2020) Land Reform and Productivity: A Quantitative Analysis with Micro Data. *American economic journal. Macroeconomics*. 12 (3), 1–39.
- Angelucci, M.** (2015) Migration and Financial Constraints: Evidence from Mexico. *The review of economics and statistics*. 97 (1), 224–228.
- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W. & Wager, S.** (2021) Synthetic Difference-in-Differences. *American Economic Review*, 111 (12), 4088–4118.
- Banerjee, A. & Lakshmi Iyer** (2005) History, Institutions, and Economic Performance: The Legacy of Colonial Land Tenure Systems in India. *The American economic review*. 95 (4), 1190–1213.
- Besley, T.** (1995) Property Rights and Investment Incentives: Theory and Evidence from Ghana. *The Journal of political economy*. 103 (5), 903–937.
- Bleakley, H. & Ferrie, J.** (2014) Land Openings on the Georgia Frontier and the Coase Theorem in the Short- and Long-Run. NBER Working Paper No. 19946. Cambridge, MA: *National Bureau of Economic Research*.
- Borşa, S.** (2013). The Sovietization of Romania. Case-study: The Collectivization of Agriculture. *Academic Journal of Interdisciplinary Studies*, 2 (8), pp. 298-303.
- Bryan, G., Chowdhury, S. & Mobarak, A. M.** (2014) Under-investment in a Profitable Technology: The Case of Seasonal Migration in Bangladesh. *NBER Working Paper Series*. 20172.
- Carter, M. R. & Barrett, C. B.** (2006) The economics of poverty traps and persistent poverty: An asset-based approach. *The Journal of development studies*. 42 (2), 178–199.
- Cartwright, A. L.** (2017) The Return of the Peasant: Land Reform in Post-Communist Romania. 1st edition. United Kingdom: *Routledge*.
- Consortium for Spatial Information (CGIAR-CSI)** (2008) SRTM 90m Digital Elevation Data. *Shuttle Radar Topography Mission*. Available at: <https://srtm.csi.cgiar.org/srtmdata/>.
- De Chaisemartin, C. and d’Haultfoeuille, X.** (2023) Credible Answers to Hard Questions: Differences-in-Differences for Natural Experiments. New Jersey: *Princeton University Press*.
- Deininger, K. & Songqing Jin** (2006) Tenure security and land-related investment: Evidence from Ethiopia. *European Economic Review*. 50 (5), 1245-1277.
- Deletant, D.** (2019) Romania under Communism: Paradox and Degeneration. London: *Routledge*.
- Dell, M.** (2010) The Persistent Effects of Peru’s Mining Mita. *Econometrica*. 78 (6), 1863–1903.
- De Soto, H.** (2001) Dead Capital and the Poor. *SAIS Review*. 21 (1), 13–44.
- Dobrinu, D.** (2004) Dinamica represiunii comuniste împotriva țăranimii din România (1951–1959). *Arhiva Istorică a României, serie nouă*. 1 (2), 253–267.

- Dustmann, C. & Görlach, J.-S.** (2016) The Economics of Temporary Migrations. *Journal of economic literature*. 54 (1), 98–136.
- Foster, A. D. & Rosenzweig, M. R.** (2022) Are There Too Many Farms in the World? Labor Market Transaction Costs, Machine Capacities, and Optimal Farm Size. *The Journal of political economy*. 130 (3), 636–680.
- Frydman, R., Gray, C., Hessel, M. & Rapaczynski, A.** (1999) When Does Privatization Work? The Impact of Private Ownership on Corporate Performance in the Transition Economies. *Quarterly Journal of Economics*, 114 (4), 1153–1191.
- Gallagher, T.** (2005) Theft of a nation: Romania since communism. *London: C. Hurst*.
- Gorton, M. & Davidova, S.** (2004) Farm productivity and efficiency in the CEE applicant countries: a synthesis of results. *Agricultural economics*. 30 (1), 1–16.
- Gow, H. R. & Swinnen, J. F. M.** (1998) Up- and downstream restructuring, foreign direct investment, and hold-up problems in agricultural transition. *European review of agricultural economics*. 25 (3), 331–350.
- Grosfeld, I. & Zhuravskaya, E.** (2015) Cultural vs. economic legacies of empires: Evidence from the partition of Poland. *Journal of Comparative Economics*. 43 (1), 55–75.
- Grossman, S. J. & Hart, O. D.** (1986) The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration. *The Journal of political economy*. 94 (4), 691–719.
- Grzymala-Busse, A.** (2008) Beyond Clientelism: Incumbent State Capture and State Formation. *Comparative political studies*. 41 (4–5), 638–673.
- Gusti, D.** (ed.) (1938) Enciclopedia României. Volumul III: Economia Națională. București, *Imprimeria Națională*.
- Hartvigsen, M.** (2014) Land reform and land fragmentation in Central and Eastern Europe. *Land Use Policy*. 36, 330–341.
- Heldring, L., Robinson, J. A. & Vollmer, S.** (2015) The Long-Run Impact of the Dissolution of the English Monasteries. Cambridge, Mass: *National Bureau of Economic Research*.
- Hirschman, A. O.** (1970) Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States. Cambridge, MA: *Harvard University Press*.
- Ioniță, N. & Mărcuță, L.** (2023) Studiu privind evoluția producției de cereale a României în perioada 2018-2022 și influența acesteia asupra rentabilității exploatațiilor agricole. *An. INCDA Fundulea*, 91 (1), p. 127-139.
- Iordachi, C. & Dobrinicu, D.** (2009) Transforming peasants, property and power: the collectivization of agriculture in Romania, 1949-1962. 1st ed. *Budapest; Central European University Press*.
- Ivănescu, S.D.** (2015). Implicarea Securității în procesul de colectivizare a agriculturii din România. *Anuarul Institutului de Istorie A.D. Xenopol - Iași*, 52, p. 383-396.
- Iyer, L. & Mani, A.** (2012) Traveling Agents: Political Change and Bureaucratic Turnover in India. *The review of economics and statistics*. 94 (3), 723–739.
- Klein, B., Crawford, R. G. & Alchian, A. A.** (1978) Vertical Integration, Appropriable Rents, and the Competitive Contracting Process. *The Journal of law & economics*. 21 (2), 297–326.
- Kligman, G. & Verdery, K.** (2011) Peasants under siege: the collectivization of Romanian agriculture, 1949-1962. *Princeton, N.J: Princeton University Press*.

- Manuelli, R. E. & Seshadri, A.** (2014) Frictionless Technology Diffusion: The Case of Tractors. *The American economic review*. 104 (4), 1368–1391.
- Mares, I. & Young, L. E.** (2019) Conditionality & coercion: electoral clientelism in Eastern Europe. *Oxford [etc: Oxford University Press]*.
- Markevich, A. & Zhuravskaya, E.** (2018) The Economic Effects of the Abolition of Serfdom: Evidence from the Russian Empire. *The American economic review*. 108 (4–5), 1074–1117.
- Mathijs, E. & Swinnen, J. F. M.** (1998) The Economics of Agricultural Decollectivization in East Central Europe and the Former Soviet Union. *Economic Development and Cultural Change*, 47 (1), 1–26.
- National Institute of Statistics (INS)** (2026) Tempo Online Database: County-level Agricultural and Demographic Indicators. *Bucharest: INSSE*. Available at: <http://statistici.insse.ro:8077/tempo-online/#/pages/tables/insse-table>.
- Nunn, N.** (2008) The Long-Term Effects of Africa's Slave Trades. Cambridge, Mass: *National Bureau of Economic Research*.
- O'Dwyer, C.** (2006) Runaway state-building: patronage politics and democratic development. 1st ed. Baltimore, Md: *Johns Hopkins University Press*.
- Permanent Electoral Authority (AEP)** (2026) Historical Electoral Results. *Data compiled by Commit Global*. Available at: <https://istoric.rezultatevot.ro/elections/60/results>.
- Pop-Eleches, G.** (2007) Historical Legacies and Post-Communist Regime Change. *The Journal of politics*. 69 (4), 908–926.
- Riley, S. J., DeGloria, S. D. & Elliot, R.** (1999) A terrain ruggedness index that quantifies topographic heterogeneity. *Intermountain Journal of Sciences*, 5 (1–4), 23–27.
- Rizov, M., Pokrivcak, J. & Ciaian, P.** (2013) CAP Subsidies and Productivity of the EU Farms. *Journal of agricultural economics*. 64 (3), 537–557.
- Roberts, H. L.** (1951) Rumania: Political Problems of an Agrarian State. *New Haven: Yale University Press*.
- Robinson, J. A. & Verdier, T.** (2013) The Political Economy of Clientelism. *The Scandinavian journal of economics*. 115 (2), 260–291.
- Sabates-Wheeler, R.** (2002). Consolidation initiatives after land reform: responses to multiple dimensions of land fragmentation in Eastern European agriculture. *Journal of International Development*, 14 (7), 1005–1018.
- Sandu, D.** (2005) Emerging Transnational Migration from Romanian Villages. *Current sociology*. 53 (4), 555–582.
- Serviciul Român de Informații** (1994) Cartea Albă a Securității (1948–1958), *București*, volume 2, p. 423–426.
- Suret-Canale, J.** (1956) Le recensement du 21 février 1956 en Roumanie. *L'information géographique*, volume 20 (4), 1956, p. 146.
- Stokes, S. C., Dunning, T., Nazareno, M. & Brusco, V.** (2013) Brokers, Voters, and Clientelism: The Puzzle of Distributive Politics. *Cambridge: Cambridge University Press*.
- Swinnen, J.F.M. & Rozelle, S.** (2006) From Marx and Mao to the market: The economics and politics of agricultural transition. *Oxford: Oxford University Press*.

- Urşanu, E.-A., Grigorescu, I., Dumitrică, C., Kucsicsa, G., Mitrică, B., Roznovietchi, I., Dumitraşcu, M. & Ciubuc, C.** (2024) Long-term changes of agricultural land over the last century in Romania. The showcase of Romanian plain. *Anthropocene*. 48, 100449.
- Verdery, K.** (2003) The vanishing hectare: property and value in postsocialist Transylvania. Ithaca, N.Y. [etc: Cornell U.P].
- Williamson, O. E.** (1987) The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting. *New York: The Free press*.
- Zimmerman, F. J. & Carter, M. R.** (2003) Asset smoothing, consumption smoothing and the reproduction of inequality under risk and subsistence constraints. *Journal of Development Economics*, 71 (2), 233–260.

Figures and Maps

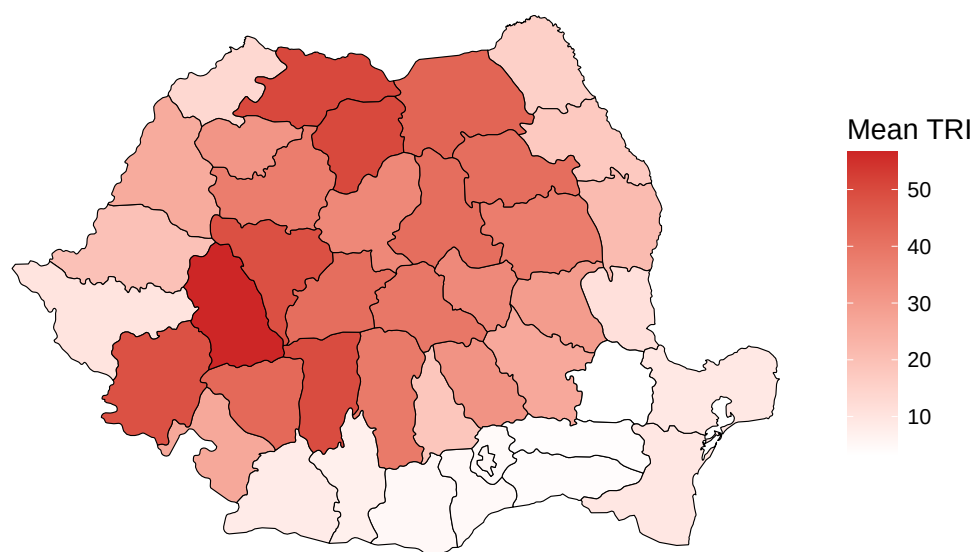


FIGURE 1: TERRAIN RUGGEDNESS INDEX ACROSS ROMANIAN COUNTIES

Notes: Larger Mean Terrain Ruggedness Index is associated with more rugged land.

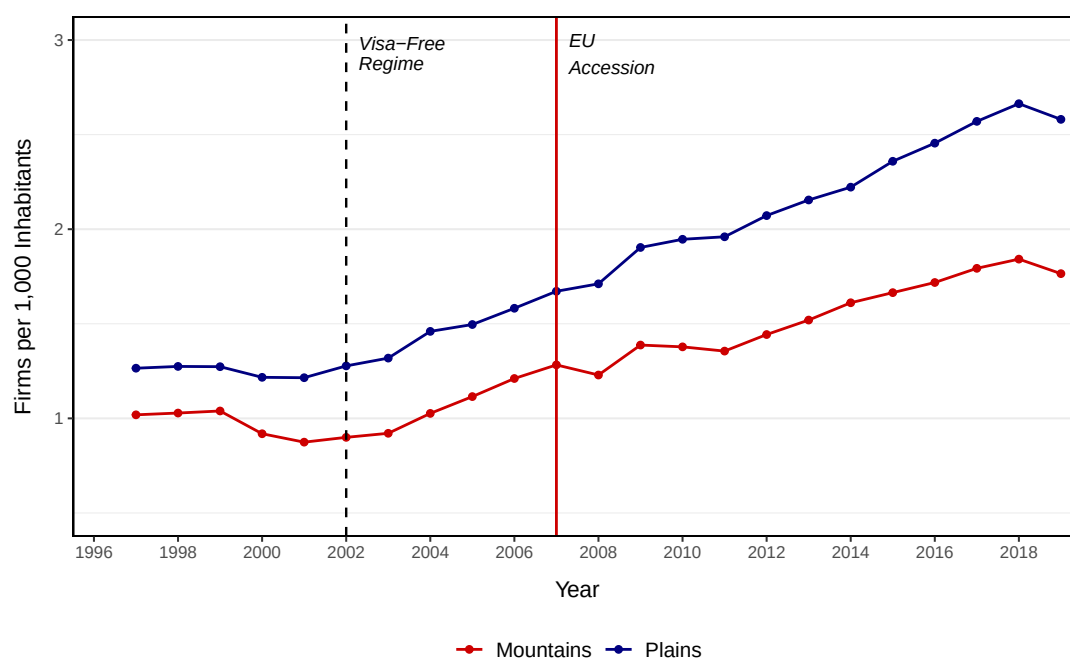


FIGURE 2: EVOLUTION OF RURAL BUSINESS FORMATION ACROSS GEOGRAPHY

Notes: Bucharest and the surrounding Ilfov county, which both belong to the Plains, are excluded from the sample.

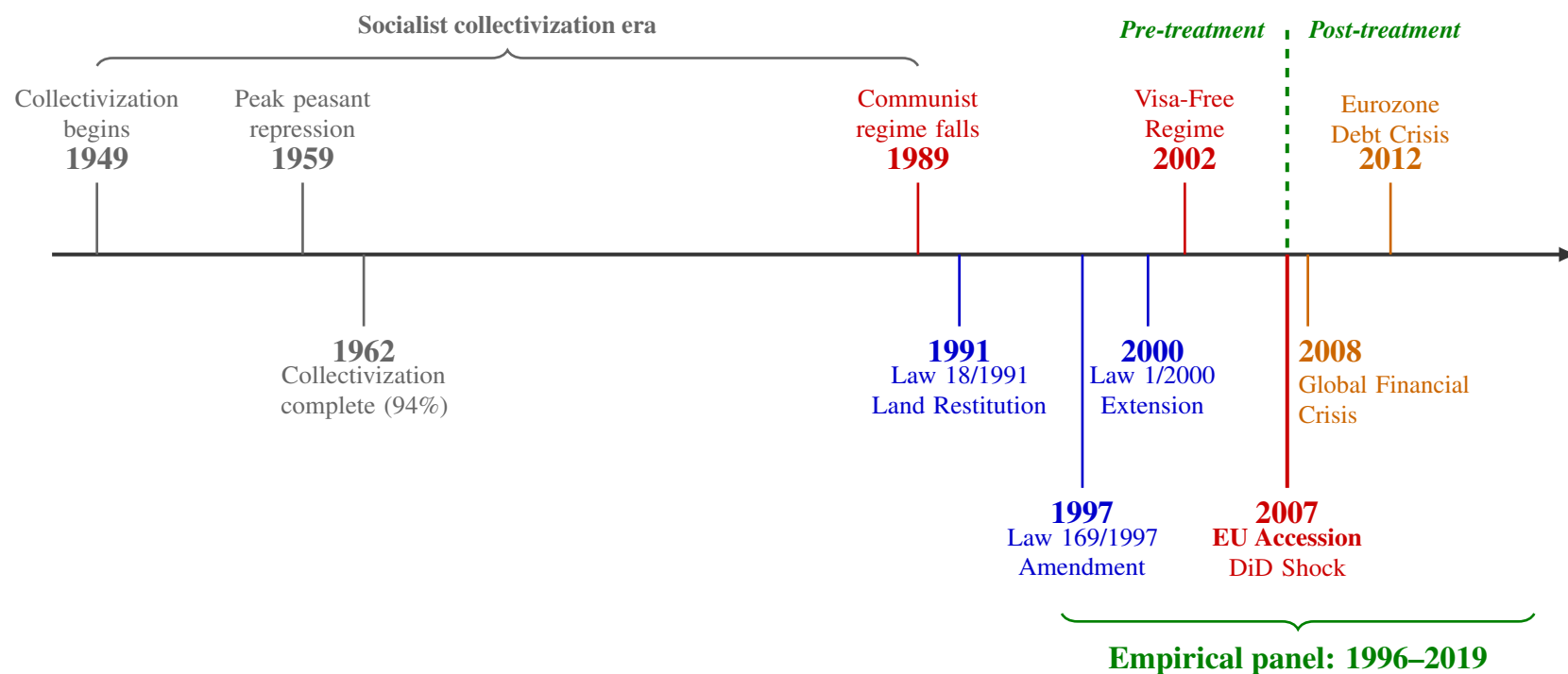


FIGURE 3: INSTITUTIONAL TIMELINE OF ROMANIAN AGRICULTURE, 1949–2019

Notes: Blue events denote domestic land restitution laws, while red and orange events highlight major structural and external macroeconomic shocks, respectively. The green bracket defines the empirical panel period used in the analysis, segmented by the 2007 EU accession treatment shock.

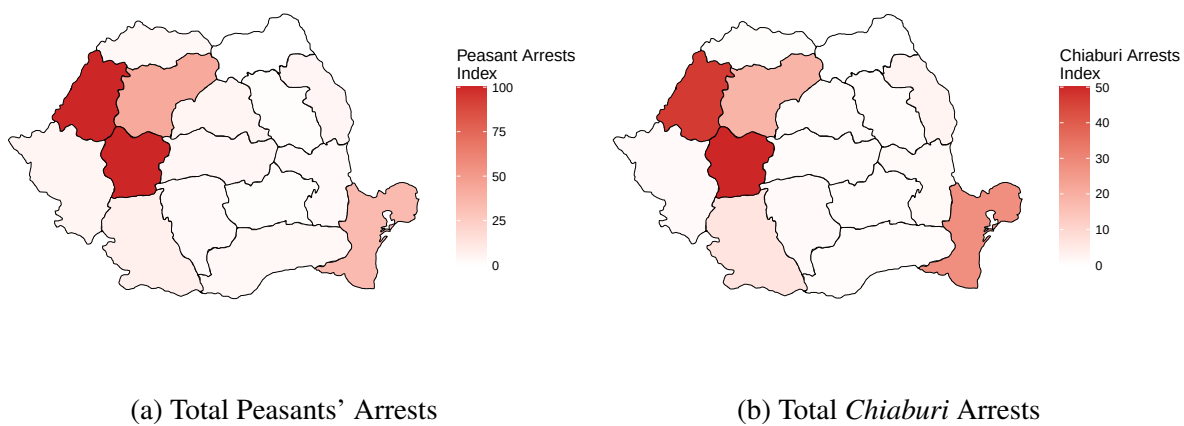


FIGURE 4: SPATIAL VARIATION IN PEASANT ARRESTS

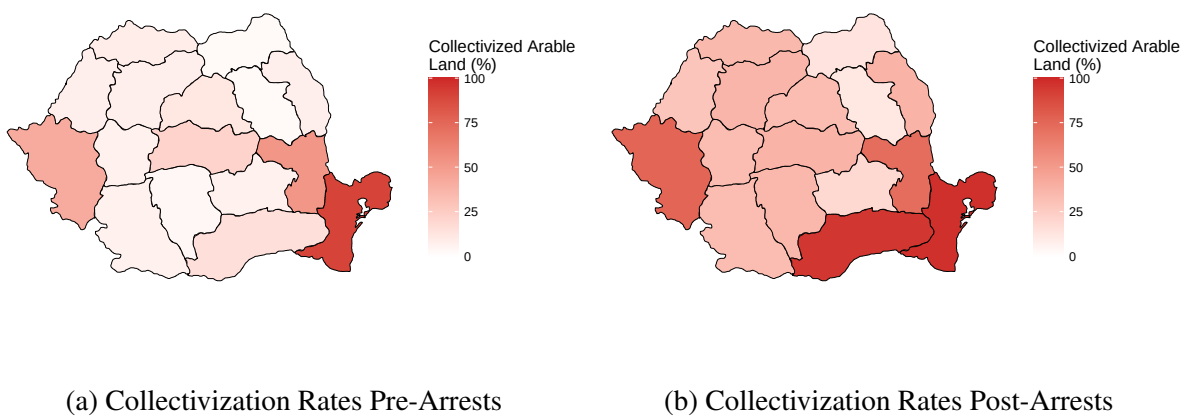


FIGURE 5: SPATIAL VARIATION IN COLLECTIVIZATION SPEED (1958–1960)

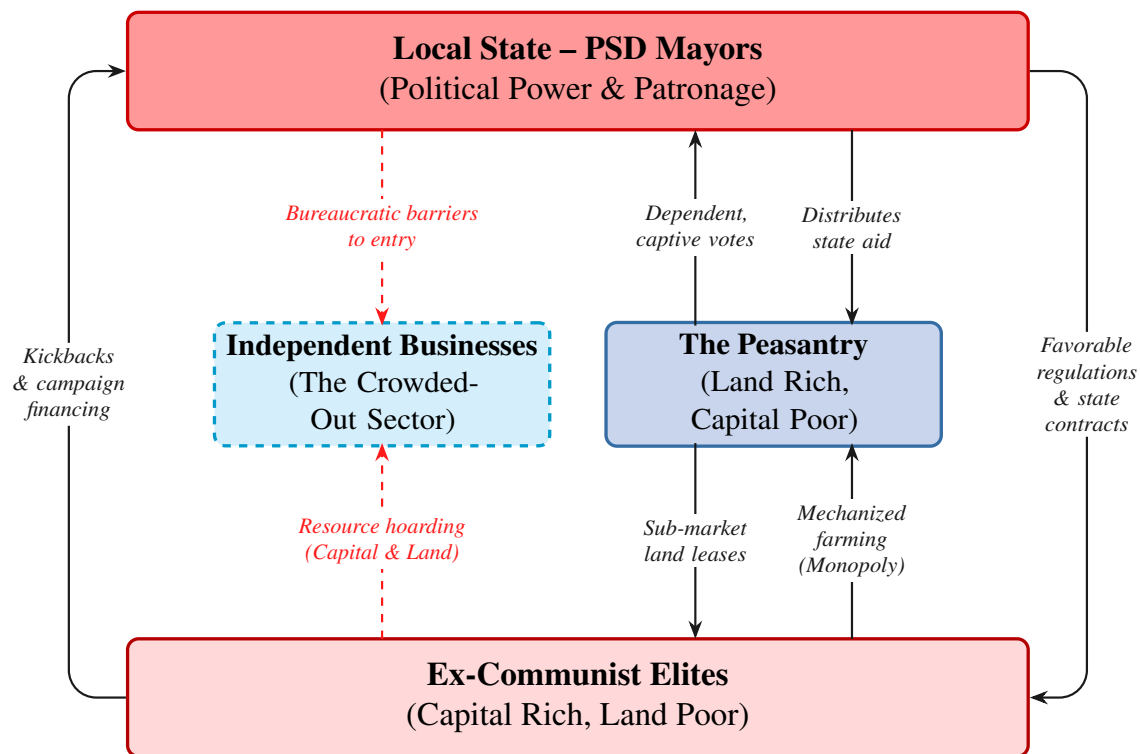


FIGURE 6: RESOURCE EXTRACTION AND BUSINESS SUPPRESSION IN THE PLAINS

Notes: The figure schematizes the Fiscal Zombie equilibrium in captured Plains counties. Solid arrows denote self-sustaining resource flows between the state, economic elites, and the dependent peasantry. Dashed red arrows indicate the structural, exclusionary pressures actively weakening the independent business sector.

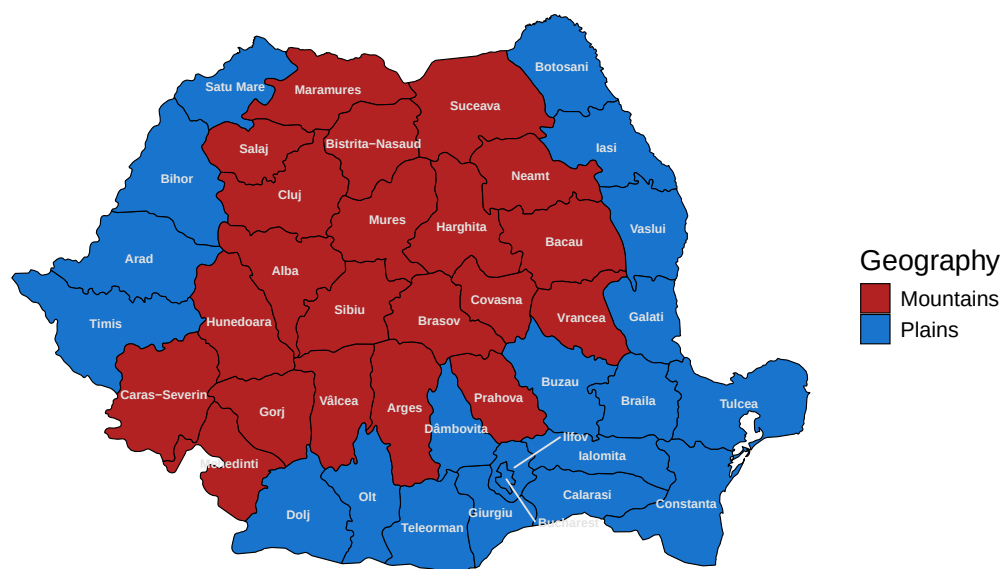


FIGURE 7: PLAINS AND MOUNTAINS DIVISION

Notes: Division is determined by the Median TRI value.

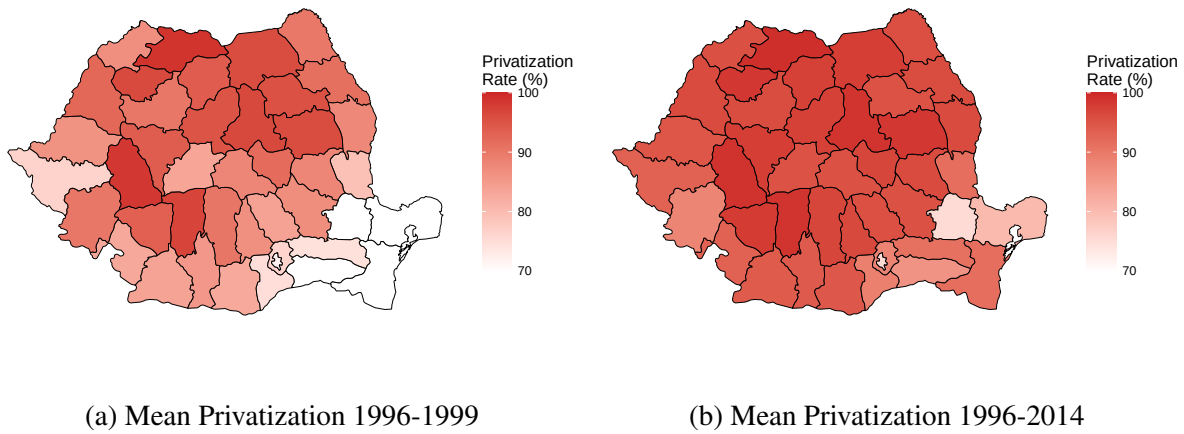


FIGURE 8: PRIVATIZATION ACROSS TIME

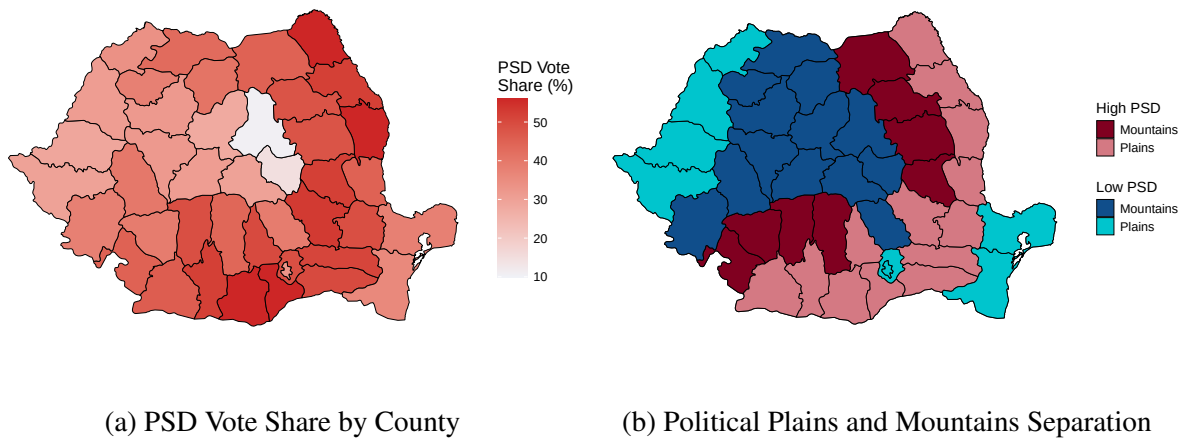


FIGURE 9: SPATIAL VARIATION OF POLITICAL CAPTURE

Notes: Division is determined by both the Median TRI and Median PSD Vote Share value.

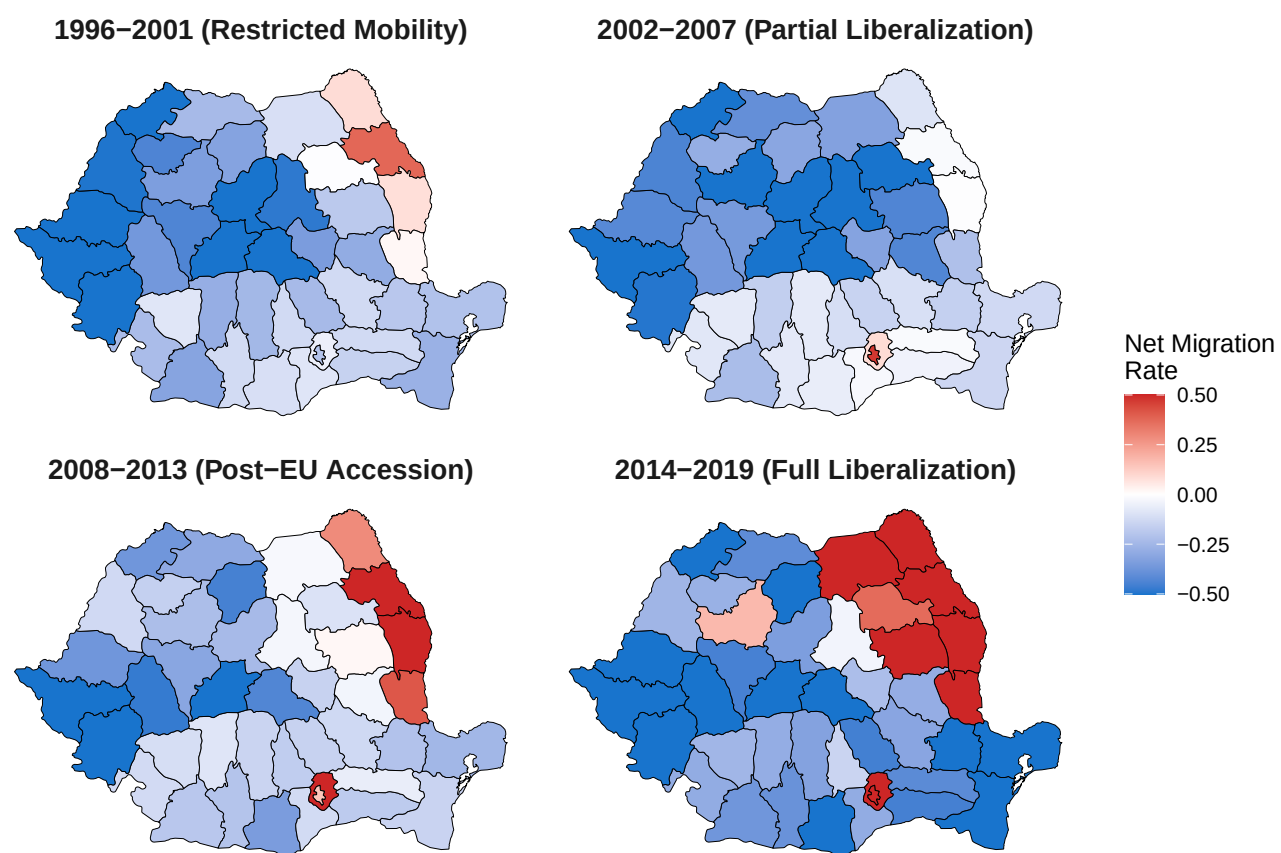


FIGURE 10: INTERNATIONAL NET MIGRATION OVER TIME

Notes: Positive values entail more immigration than emigration and vice versa.

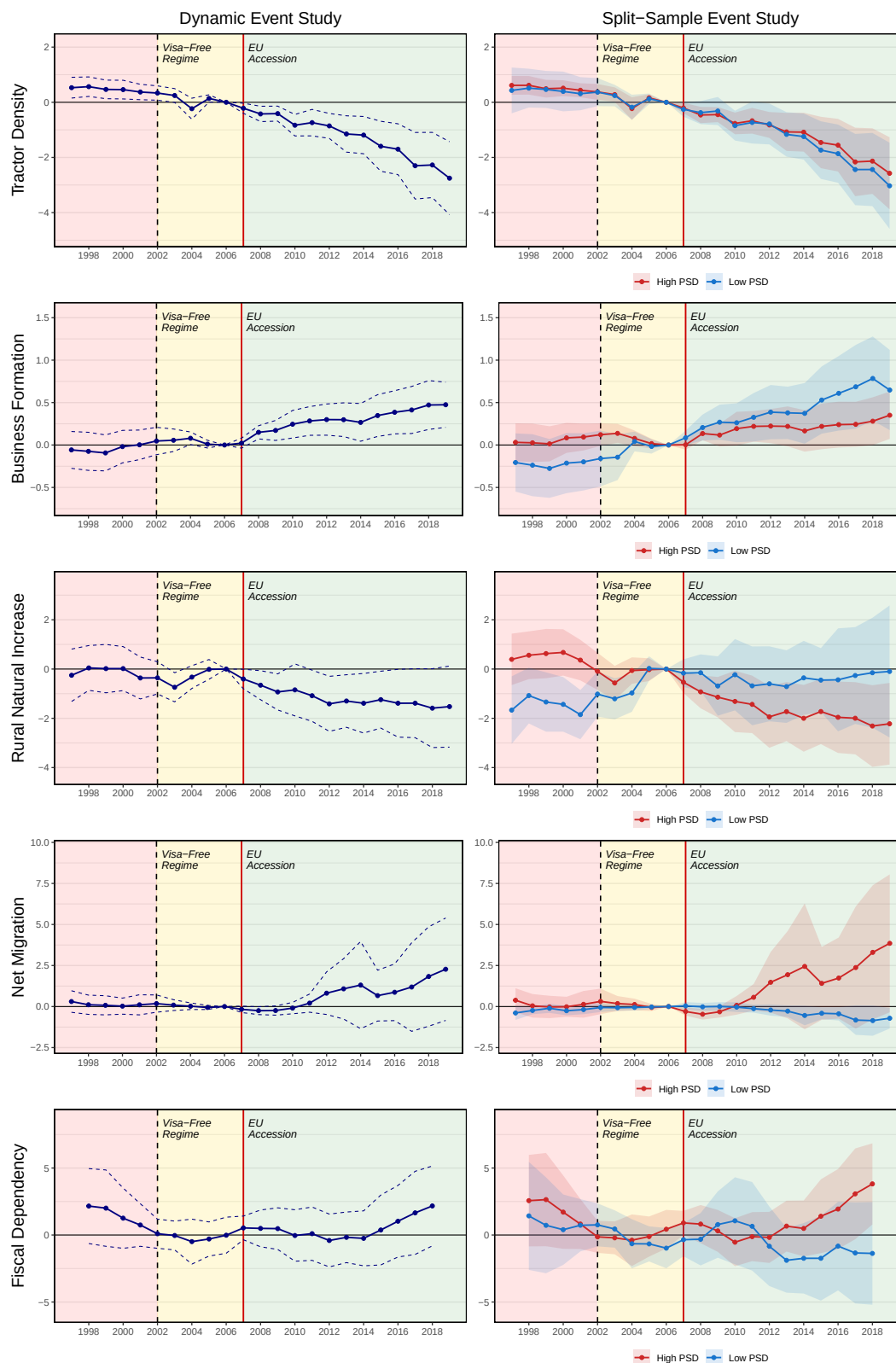


FIGURE 11: DYNAMIC & SPLIT-SAMPLE EVENT STUDIES

Notes: Left column: aggregate event study estimates (Equation 4). Right column: split-sample estimates partitioning the Plains into High- and Low-PSD cohorts. Solid lines and markers denote point estimates (β_k), with 95% confidence intervals shown as dashed lines (left) and shaded regions (right). To filter short-term volatility, Fiscal Dependency is smoothed via a 3-year centered moving average.

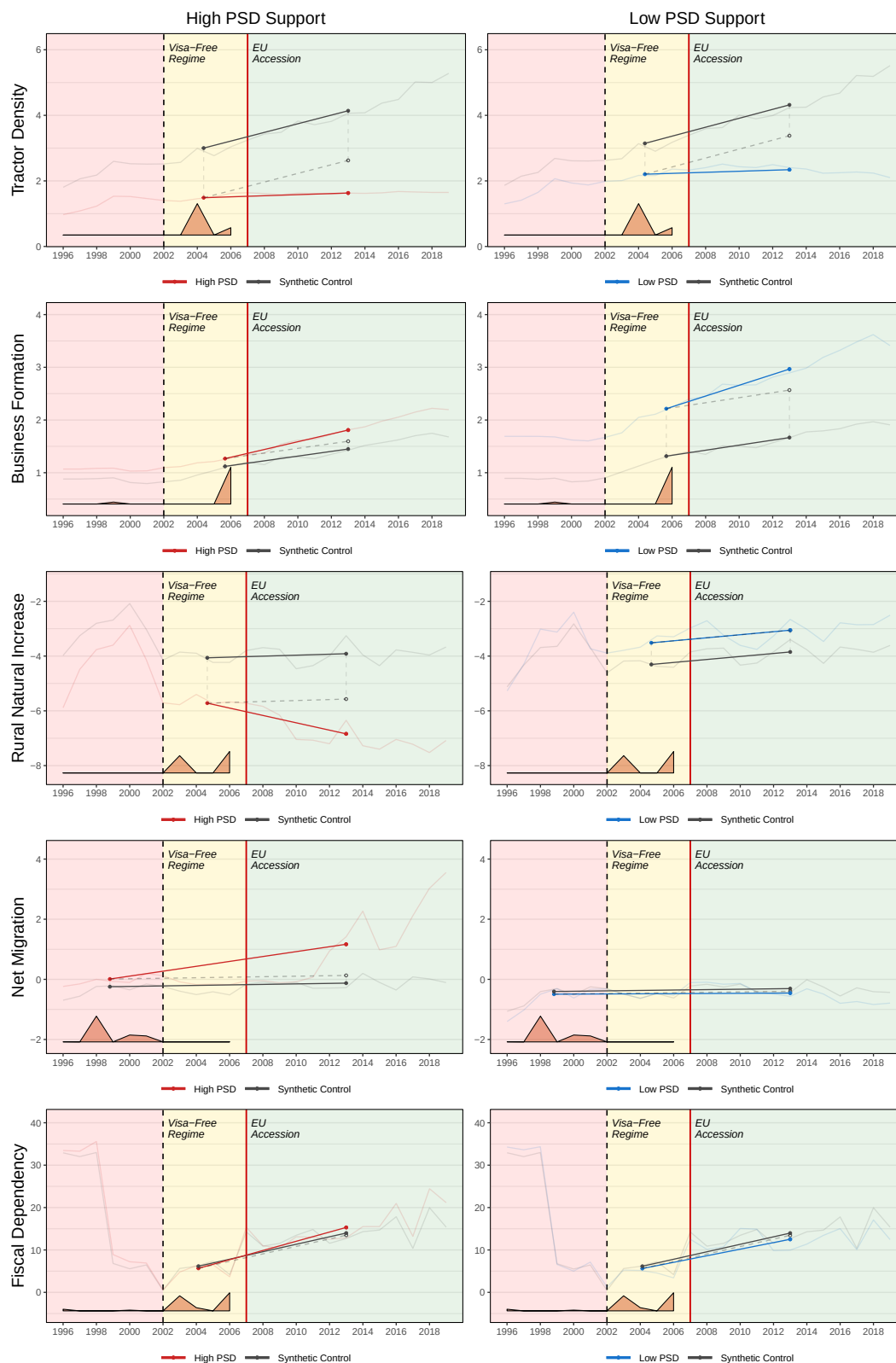


FIGURE 12: SYNTHETIC DID TRAJECTORIES

Notes: These figures plot the observed actual trajectories of the treated Plains sub-cohorts against their respective synthetic controls constructed via the SDiD estimator (Equation 5). The left column evaluates the High-PSD sample, while the right column evaluates the Low-PSD sample as a falsification test.

Tables

TABLE 1: HISTORICAL ARRESTS AND COLLECTIVIZATION (1959–1960)

	Obs.	Mean	Std. Dev	Min	Median	Max
Panel A: Full Sample Averages						
Peasant Arrests Index	16	50.79	135.85	0.05	4.40	538.79
Chiaburi Arrests Index	16	37.01	118.94	0.05	1.29	480.33
Collectivization Rate	16	43.56	26.61	12.00	35.50	97.00
	Plains (TRI < Median)			Mountains (TRI > Median)		
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev
Panel B: Comparison by Geography						
Peasant Arrests Index	8	26.90	53.50	8	74.70	188.00
Chiaburi Arrests Index	8	11.10	17.20	8	63.00	169.00
Collectivization Rate	8	57.40	31.20	8	29.80	10.50
Panel C: Regional Classification by TRI Score						
Plains (TRI < Median)				Mountains (TRI > Median)		
Region	TRI Score		Region	TRI Score		
Bucharest	4.50		Suceava	32.80		
Dobrogea	9.10		Maramureş	33.10		
Galaţi	14.20		Braşov	36.00		
Iaşi	19.70		Argeş	36.60		
Oltenia	20.30		Bacău	37.80		
Crişana	22.70		Maghiară	39.90		
Banat	26.60		Cluj	44.50		
Ploieşti	31.40		Hunedoara	54.10		

Notes: Panel A reports peasants and *chiaburi* arrests indices that refer to the year 1959, whereas collectivization rate data refer to the year 1960. Panel B divides the sample by the median Terrain Ruggedness Index (TRI). Panel C lists the 16 historical regions sorted by their mean TRI. 'Plains' indicates mean TRI below the sample median value; 'Mountains' indicates TRI above the median value.

TABLE 2: TESTABLE IMPLICATIONS OF THE CONCEPTUAL FRAMEWORK

Outcome Variable	Unconditional Plains Effect	Conditional on High PSD Capture	Primary Timing
<i>Asset Mismatch Channel</i>			
Privatization Rate	—	<i>Not Tested</i>	1996–1999
Tractor Density	—	—	Post-2007
<i>Immobility & Political Capture Channel</i>			
Rural Business Formation	+ / 0	— / 0	Post-2007
External Migration	±	±	Post-2007
Rural Natural Increase	— / 0	—	Post-2007
Fiscal Dependency	0	+	Post-2007

Notes: — and + denote the predicted direction of the Plains dummy coefficient. Double signs (—, ++) indicate a stronger predicted magnitude conditional on high PSD vote share. 0 indicates no predicted unconditional level effect. ± indicates theoretical ambiguity in sign. PSD Vote Share serves as the moderating variable in split-sample and synthetic DiD specifications rather than as an outcome variable. Timing refers to the period in which divergence is predicted to be most pronounced relative to the 2007 EU accession shock.

TABLE 3: DESCRIPTIVE STATISTICS

	Obs.	Mean	Std. Dev	Min	Median	Max	
Panel A: Cleaned Sample							
<i>Core Variables</i>							
Privatization Rate (1996–1999)	160	86.60	10.40	58.60	88.20	99.30	
Privatization Rate (1996–2014)	760	94.40	7.38	58.60	97.50	99.80	
Tractor Density	960	2.84	2.16	0.46	2.16	15.58	
<i>Modern Variables</i>							
Rural Business Formation	920	1.55	0.85	0.33	1.35	4.79	
Net Migration Rate	960	-0.03	1.82	-3.25	-0.23	23.01	
Rural Natural Increase	959	-4.41	3.76	-14.45	-4.35	5.86	
Fiscal Dependency Ratio	960	13.60	9.54	0.04	11.54	47.49	
PSD Vote Share	200	35.09	14.70	4.00	34.50	69.00	
<i>Control Variables</i>							
Modernized Roads Ratio	960	57.94	10.87	33.45	57.37	86.04	
High School Graduates Ratio	960	10.63	2.79	4.10	10.34	20.57	
Hospital Beds Index	920	5.84	1.58	2.62	5.58	11.11	
Social Aid Index	960	32.80	30.66	0.48	23.80	237.13	
	Plains (TRI < Median)			Mountains (TRI > Median)			t-test
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev	Value
Panel B: Comparison by TRI Score							
<i>Core Variables</i>							
Privatization Rate (1996–1999)	76	80.10	9.91	84	92.40	4.83	4.89***
Privatization Rate (1996–2014)	361	92.00	5.71	399	96.60	2.48	3.28***
Tractor Density	456	1.71	0.78	504	3.86	1.71	5.21***
<i>Modern Variables</i>							
Rural Business Formation	437	1.81	0.83	483	1.31	0.52	-2.28**
Net Migration Rate	456	0.24	1.37	504	-0.29	0.38	-1.64
Rural Natural Increase	456	-5.09	3.69	503	-3.78	3.35	1.17
Fiscal Dependency Ratio	456	13.90	1.74	504	13.40	2.20	-0.69
PSD Vote Share	95	39.70	9.74	105	30.90	11.60	-2.49**
<i>Control Variables</i>							
Modernized Roads Ratio	456	59.30	9.34	504	56.70	8.02	-0.93
High School Graduates Ratio	456	10.10	1.17	504	11.10	1.02	3.02***
Hospital Beds Index	437	5.50	1.42	483	6.15	1.39	1.45
Social Aid Index	456	26.40	4.90	504	38.70	13.40	3.94***

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Panel A reports summary statistics for the cleaned sample (1996–2019). Panel B divides the sample by the median Terrain Ruggedness Index (TRI). 'Plains' indicates mean TRI below the sample median; 'Mountains' indicates mean TRI above the median.

TABLE 4: OLS REGRESSION RESULTS

Dependent Variable:	(1) Privatization Rate	(2) Tractor Density	(3) Business Formation	(4) External Migration	(5) Rural Natural Increase	(6) Fiscal Dependency	(7) PSD Vote Share
Plains Dummy	-3.377*** (1.132)	-1.783*** (0.417)	0.319 (0.208)	0.354** (0.154)	-0.288 (1.211)	-0.452 (0.525)	7.061** (2.721)
Privatization Rate		-0.008 (0.028)	-0.022** (0.011)	0.007 (0.008)	0.082 (0.058)	-0.064 (0.044)	-0.201 (0.186)
Dependent Variable Mean	94.399 (7.378)	2.589 (1.631)	1.386 (0.744)	-0.180 (1.078)	-4.301 (3.777)	12.892 (10.115)	32.125 (13.075)
Observations	760	760	720	760	759	760	160
Adjusted R^2	0.535	0.542	0.436	0.140	0.251	0.911	0.554
Adjusted Within R^2	0.336	0.502	0.360	0.086	0.244	0.085	0.458
Controls	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County
No. of Counties	40	40	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE 5: POOLED DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Plains $\times$ Post-2007	-1.556*** (0.331)	0.300** (0.115)	0.666 (0.537)	-0.976 (0.598)	0.007 (0.730)
Dependent Variable Mean	2.898 (2.079)	1.546 (0.851)	-0.005 (1.846)	-4.385 (3.751)	12.785 (8.773)
Observations	920	920	920	919	920
Adjusted R^2	0.822	0.921	0.337	0.899	0.876
Adjusted Within R^2	0.185	0.155	0.038	0.073	0.056
Controls	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include County and Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, Hospital Beds Index, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE 6: SPLIT-SAMPLE DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	Tractor Density		Business Formation		External Migration		Rural Natural Increase		Fiscal Dependency	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)	High (7)	Low (8)	High (9)	Low (10)
Plains × Post-2007	-1.497*** (0.293)	-1.585*** (0.485)	0.140 (0.126)	0.564*** (0.136)	1.282 (0.788)	-0.208 (0.155)	-1.801*** (0.616)	0.642 (0.682)	0.552 (0.823)	-1.214 (0.917)
Dependent Variable Mean	3.025 (2.189)	3.552 (2.207)	1.384 (0.733)	1.563 (0.843)	0.080 (1.984)	-0.312 (0.652)	-4.590 (3.889)	-3.646 (3.306)	12.976 (8.811)	12.391 (8.509)
Observations	782	621	782	621	782	621	781	620	782	621
Adjusted R^2	0.825	0.783	0.912	0.933	0.365	0.315	0.918	0.895	0.877	0.881
Adjusted Within R^2	0.188	0.093	0.092	0.315	0.077	0.010	0.168	0.127	0.064	0.041
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	34	27	34	27	34	27	34	27	34	27

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include County and Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, Hospital Beds Index, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE 7: SYNTHETIC DIFFERENCE-IN-DIFFERENCES RESULTS

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
SDiD Estimate ($\hat{\tau}^{SDiD}$)	-0.994*** (0.220)	0.213** (0.094)	1.037 (0.743)	-1.270** (0.513)	1.829** (0.834)
Observations	816	782	816	816	816
Treated Counties	13	13	13	13	13
Donor Pool (Mountains)	21	21	21	21	21
Time Weights ($\hat{\lambda}_t$)	Estimated	Estimated	Estimated	Estimated	Estimated
Unit Weights ($\hat{\omega}_c$)	Estimated	Estimated	Estimated	Estimated	Estimated
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Standard Errors	Jackknife	Jackknife	Jackknife	Jackknife	Jackknife

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Jackknife standard errors are reported in parentheses. All models include County and Time Fixed Effects. The model matches High-PSD Plains to a synthetic Mountains baseline. Dependent variable means are omitted given that the SDiD estimator evaluates effects relative to a weighted synthetic baseline rather than an unweighted sample average.

Appendix

A. Variable Construction and Sources

This section provides supplementary descriptive statistics and classification thresholds for the analytical sample. [Table A1](#) details the specific mean terrain ruggedness indices for each county, [Tables A2](#) and [A3](#) report full pre-processed summary statistics and processed statistics based on political capture, respectively. Overall, they quantify the pre-existing structural and institutional disparities between the Plains and the Mountains as well as within the Plains prior to the application of the identification strategy.

TABLE A1: CLASSIFICATION OF COUNTIES BY TERRAIN RUGGEDNESS INDEX (TRI)

Plains (TRI < Median)		Mountains (TRI > Median)	
County	TRI Score	County	TRI Score
Brăila	3.20	Mehedinți	26.50
Ialomița	3.60	Vrancea	29.60
Călărași	3.90	Sălaj	31.40
Ilfov	4.60	Prahova	31.90
Giurgiu	4.90	Covasna	34.50
Teleorman	5.00	Mureș	34.80
Bucharest	5.10	Bacău	37.60
Olt	7.10	Cluj	37.60
Dolj	8.40	Argeș	38.60
Tulcea	9.20	Brașov	39.30
Constanța	9.50	Sibiu	41.50
Timiș	10.30	Harghita	41.50
Galați	11.30	Neamț	41.60
Satu Mare	13.50	Gorj	42.70
Botoșani	15.60	Suceava	43.80
Iași	17.50	Caraș-Severin	48.40
Dâmbovița	18.50	Alba	48.50
Arad	19.60	Vâlcea	49.50
Vaslui	21.20	Bistrița-Năsăud	50.30
Bihor	25.70	Maramureș	50.30
Buzău	26.30	Hunedoara	56.50

Notes: The sample is split at the median TRI value.

TABLE A2: DESCRIPTIVE STATISTICS

	Obs.	Mean	Std. Dev	Min	Median	Max
Panel A: Full Sample						
<i>Core Variables</i>						
Privatization Rate (1996–1999)	166	86.14	10.43	58.64	87.92	99.25
Privatization Rate (1996–2014)	796	93.85	7.86	58.64	97.29	99.78
Tractor Density	1,004	3.92	16.31	0.46	2.18	483.33
<i>Modern Variables</i>						
Rural Business Formation	943	1.54	0.85	0.33	1.33	4.79
Net Migration Rate	1,008	-0.003	1.81	-3.25	-0.22	23.01
Rural Natural Increase	983	-4.34	3.74	-14.45	-4.26	5.86
Fiscal Dependency Ratio	1,008	13.40	9.71	0.04	11.30	51.50
PSD Vote Share	210	34.97	11.28	6.40	36.80	50.40
<i>Control Variables</i>						
Modernized Roads Ratio	1,008	59.60	13.07	33.45	58.72	100.00
High School Graduates Ratio	1,008	10.54	2.85	3.59	10.27	20.57
Hospital Beds Index	966	5.91	1.68	2.62	5.62	11.11
Social Aid Index	1,006	32.19	30.25	0.48	23.32	237.13
	Plains (TRI < Median)			Mountains (TRI > Median)		
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev
Panel B: Comparison by TRI Score						
<i>Core Variables</i>						
Privatization Rate (1996–1999)	82	79.6	9.59	84	92.4	4.83
Privatization Rate (1996–2014)	397	91.0	6.54	399	96.6	2.48
Tractor Density	500	4.39	12.2	504	3.86	1.71
<i>Modern Variables</i>						
Rural Business Formation	460	1.78	0.82	483	1.31	0.52
Net Migration Rate	504	0.28	1.30	504	-0.29	0.38
Rural Natural Increase	480	-4.93	3.66	503	-3.78	3.35
Fiscal Dependency Ratio	504	13.30	2.43	504	13.40	2.20
PSD Vote Share	105	39.10	9.51	105	30.90	11.60
<i>Control Variables</i>						
Modernized Roads Ratio	504	62.40	13.50	504	56.70	8.02
High School Graduates Ratio	504	9.95	1.40	504	11.10	1.02
Hospital Beds Index	483	5.68	1.66	483	6.15	1.39
Social Aid Index	502	25.60	5.17	504	38.70	13.40

Notes: Panel A reports summary statistics for the full sample (1996–2019). Panel B divides the sample by the median Terrain Ruggedness Index (TRI). 'Plains' indicates mean TRI below the sample median; 'Mountains' indicates mean TRI above the median.

TABLE A3: DESCRIPTIVE STATISTICS: PLAINS COHORTS BY PSD VOTE SHARE

	High-PSD Plains			Low-PSD Plains			t-test
	Obs.	Mean	Std. Dev	Obs.	Mean	Std. Dev	Value
<i>Core Variables</i>							
Privatization Rate (1996–1999)	52	80.60	9.66	24	79.00	11.30	0.31
Privatization Rate (1996–2014)	247	92.00	5.87	114	92.00	5.90	0.01
Tractor Density	312	1.52	0.61	144	2.14	0.98	-1.44
<i>Modern Variables</i>							
Rural Business Formation	299	1.51	0.75	138	2.46	0.60	-2.98**
Net Migration Rate	312	0.60	1.53	144	-0.52	0.25	2.57**
Rural Natural Increase	312	-5.92	3.94	144	-3.31	2.48	-1.75*
Fiscal Dependency Ratio	312	14.40	1.68	144	12.60	1.24	2.60**
PSD Vote Share	65	42.20	3.21	30	25.50	6.63	5.87***
<i>Control Variables</i>							
Modernized Roads Ratio	312	59.20	11.10	144	59.50	4.22	-0.08
High School Graduates Ratio	312	9.81	1.26	144	10.60	0.76	-1.8*
Hospital Beds Index	299	5.24	1.37	138	6.08	1.45	-1.20
Social Aid Index	312	27.10	4.85	144	24.80	5.06	0.94

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports summary statistics exclusively for counties classified as Plains (TRI below the full 42 counties median). The sample is further divided by the median average PSD vote share (1996–2016) of the Plains cohort. 'High-PSD Plains' indicates an average vote share above the cohort median; 'Low-PSD Plains' indicates a mean vote share below the median.

B. Spatial Variation of Key Variables

This section visualizes the spatial distribution of the primary economic, demographic, and fiscal outcome variables that were not explicitly displayed in the main body of the paper. [Figures B1, B2, B3, and B4](#) map the county-level variation in tractor density, rural business formation, rural natural increase, and fiscal dependency, respectively. The former is also divided into four different historical periods that defined the evolution of demographic tendencies. Apart from fiscal dependency, which shows ambiguous spatial disparities, the maps visually corroborate the empirical results by illustrating the geographic clustering of agricultural decay and the spatial footprint of the demographic trap as well as political capture.

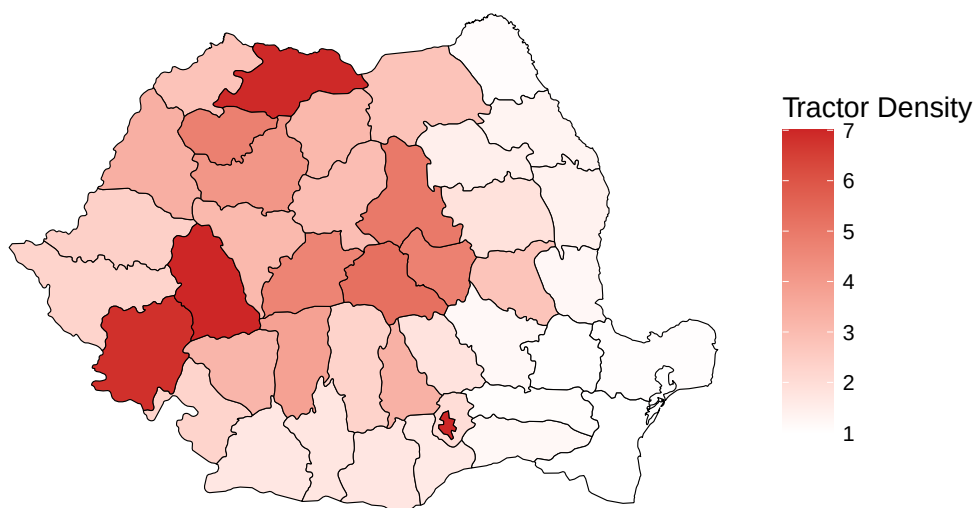


FIGURE B1: SPATIAL VARIATION OF TRACTOR DENSITY

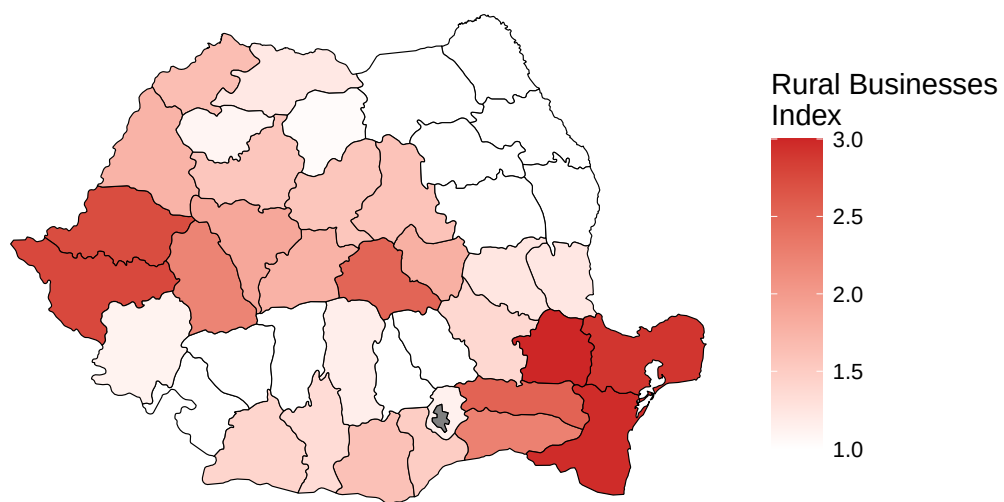


FIGURE B2: SPATIAL VARIATION OF RURAL BUSINESS FORMATION

Notes: Bucharest Municipality does not have rural population.

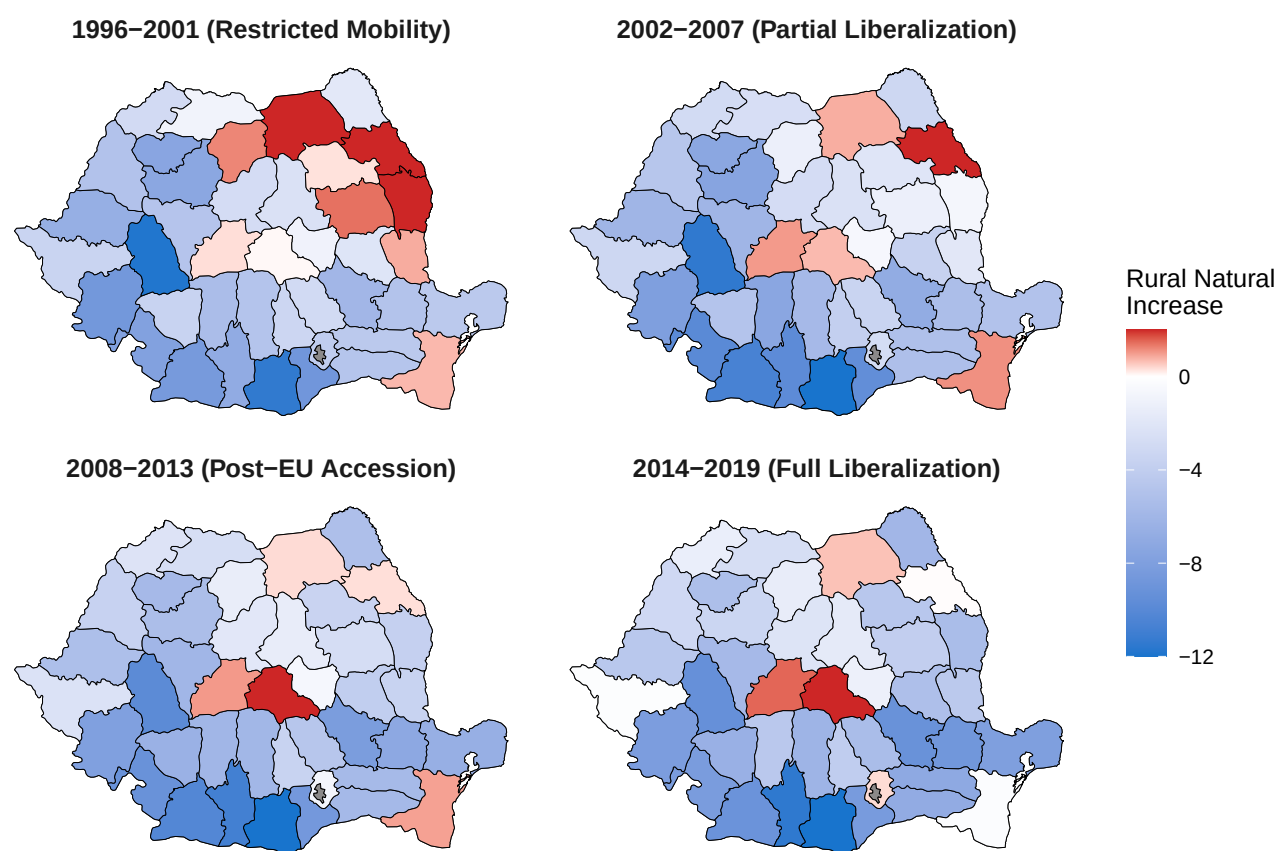


FIGURE B3: RURAL NATURAL INCREASE OVER TIME

Notes: Positive values entail more births than deaths and vice versa. Bucharest Municipality does not have rural population.

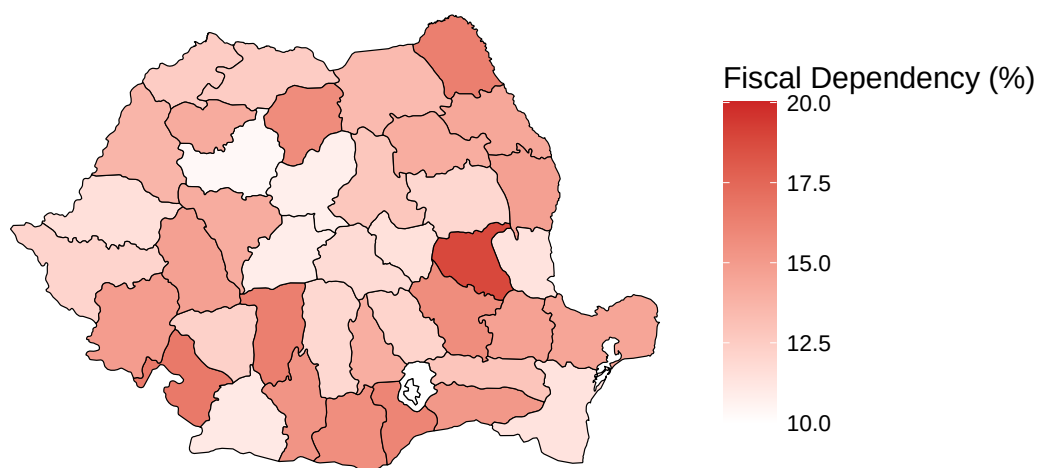


FIGURE B4: SPATIAL VARIATION OF FISCAL DEPENDENCY

C. Identification Without Controls

This section replicates the primary empirical models without the inclusion of socioeconomic controls. [Tables C1, C2, C3, C4, C5, and C6](#) report these unadjusted estimates for the OLS, baseline DiD, split-sample DiD, and three robustness checks (their full model results can be found in [Appendix D](#)), respectively. They are alongside [Figure C1](#), which visualizes baseline and split-sample DiD event studies without controls. Comparing all of these outputs to the fully specified models presented in the main text reveals no major shifts in the direction of the estimates or any other discrepancies, demonstrating that the core geographic and political treatment effects are plausibly driven mainly by the external shocks rather than being artifacts of endogenous post-treatment conditioning.

TABLE C1: OLS REGRESSION RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	(1) Privatization Rate	(2) Tractor Density	(3) Business Formation	(4) External Migration	(5) Rural Natural Increase	(6) Fiscal Dependency	(7) PSD Vote Share
Plains Dummy	-4.745*** (1.356)	-1.790*** (0.432)	0.193 (0.211)	0.377** (0.190)	-1.338 (1.221)	0.091 (0.568)	8.054** (3.455)
Privatization Rate		-0.006 (0.028)	-0.036** (0.014)	0.015 (0.009)	0.053 (0.050)	-0.037 (0.035)	-0.088 (0.161)
Dependent Variable Mean	94.399 (7.378)	2.589 (1.631)	1.386 (0.744)	-0.180 (1.078)	-4.301 (3.777)	12.892 (10.115)	32.125 (13.075)
Observations	760	760	720	760	759	760	160
Adjusted R^2	0.404	0.378	0.244	0.083	0.059	0.903	0.276
Adjusted Within R^2	0.150	0.323	0.142	0.026	0.049	0.004	0.121
Controls							
Time FE	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County
No. of Counties	40	40	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE C2: POOLED DIFFERENCE-IN-DIFFERENCES RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Plains × Post-2007	-1.613*** (0.334)	0.304** (0.119)	0.619 (0.527)	-0.980 (0.602)	-0.031 (0.916)
Dependent Variable Mean	2.842 (2.059)	1.546 (0.851)	-0.034 (1.818)	-4.407 (3.756)	13.638 (9.541)
Observations	960	920	960	959	960
Adjusted R^2	0.815	0.915	0.314	0.891	0.888
Adjusted Within R^2	0.179	0.089	0.010	0.039	-0.001
Controls					
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include County and Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE C3: SPLIT-SAMPLE DIFFERENCE-IN-DIFFERENCES RESULTS: MODEL WITHOUT CONTROLS

Dependent Variable:	Tractor Density		Business Formation		External Migration		Rural Natural Increase		Fiscal Dependency	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)	High (7)	Low (8)	High (9)	Low (10)
Plains × Post-2007	-1.672*** (0.335)	-1.484*** (0.416)	0.157 (0.131)	0.622*** (0.147)	0.966 (0.742)	-0.133 (0.161)	-1.794*** (0.590)	0.783 (0.771)	0.664 (1.047)	-1.537 (0.998)
Dependent Variable Mean	2.965 (2.168)	3.480 (2.194)	1.384 (0.733)	1.563 (0.843)	0.052 (1.950)	-0.338 (0.669)	-4.601 (3.895)	-3.680 (3.322)	13.815 (9.544)	13.257 (9.332)
Observations	816	648	782	621	816	648	815	647	816	648
Adjusted R^2	0.816	0.778	0.905	0.927	0.322	0.340	0.912	0.876	0.887	0.898
Adjusted Within R^2	0.168	0.086	0.028	0.254	0.021	0.001	0.131	0.019	0.001	0.011
Controls										
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	34	27	34	27	34	27	34	27	34	27

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Standard errors are clustered at the county level. All models include County and Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

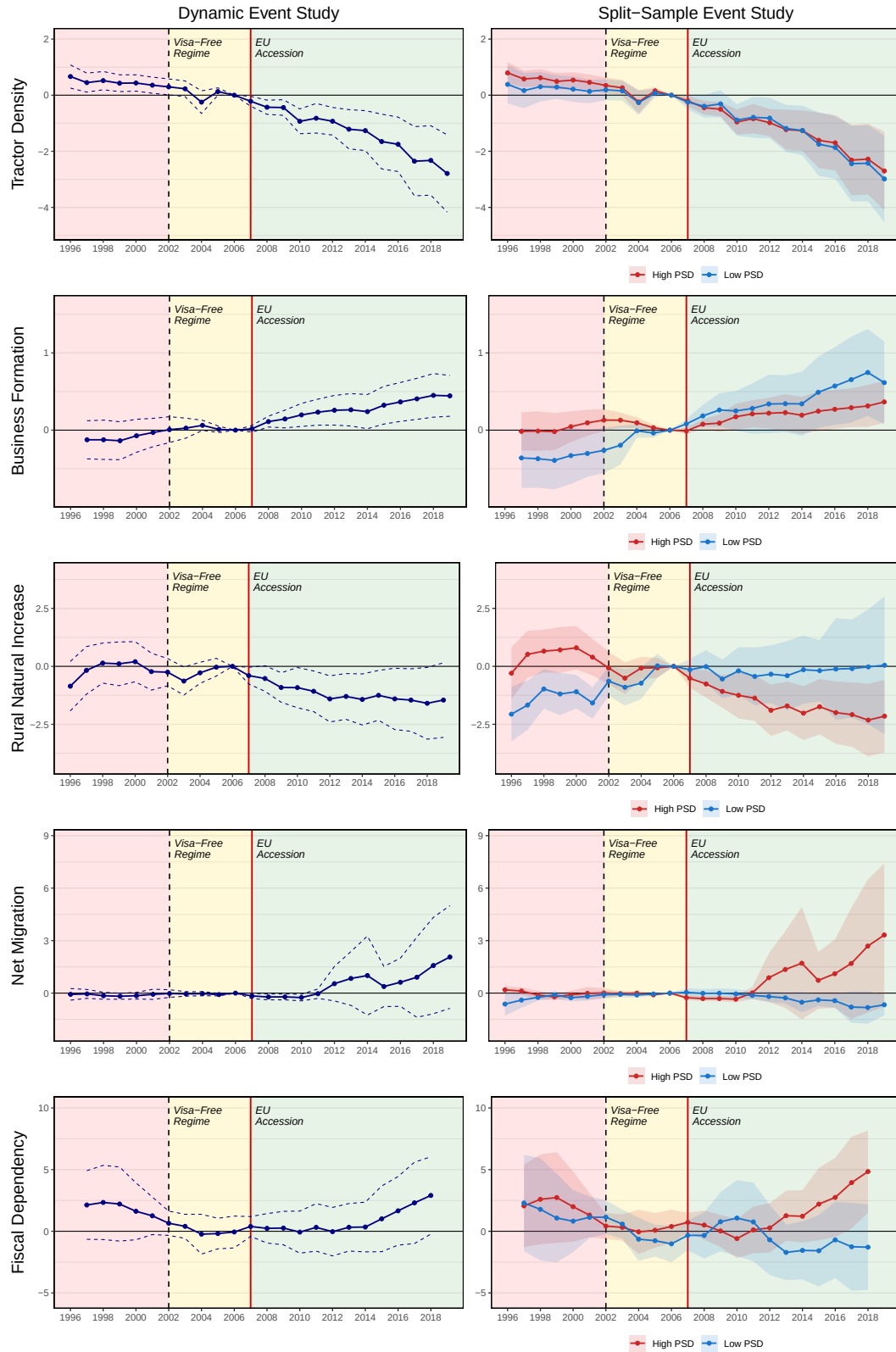


FIGURE C1: DYNAMIC & SPLIT-SAMPLE EVENT STUDIES: MODELS WITHOUT CONTROLS

Notes: Left column: aggregate event study estimates (Equation 4, but without controls). Right column: split-sample estimates partitioning the Plains into High- and Low-PSD cohorts. Solid lines and markers denote point estimates (β_k), with 95% confidence intervals shown as dashed lines (left) and shaded regions (right). To filter short-term volatility, Fiscal Dependency is smoothed via a 3-year centered moving average.

TABLE C4: ROBUSTNESS CHECK: EXTREME TRI SPLIT WITHOUT CONTROLS

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Extreme Plains $\times$ Post-2007	-2.463*** (0.547)	0.411** (0.183)	-0.378 (0.231)	-0.428 (0.719)	-0.146 (1.425)
Dependent Variable Mean	2.893 (2.447)	1.748 (0.953)	-0.267 (0.674)	-5.449 (3.983)	14.000 (9.588)
Observations	480	460	480	479	480
Adjusted R^2	0.821	0.916	0.253	0.920	0.887
Adjusted Within R^2	0.279	0.131	0.028	0.010	0.000
Controls					
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	20	20	20	20	20

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This sample restricts the dataset to the top 25% most rugged (Mountains) and bottom 25% flattest (Plains) counties. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize baseline DiD framework and include County and Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE C5: ROBUSTNESS CHECK: EXTREME PSD PLAINS SPLIT WITHOUT CONTROLS

Dependent Variable:	Tractor Density		Business Formation		External Migration		Rural Natural Increase		Fiscal Dependency	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)	High (7)	Low (8)	High (9)	Low (10)
PSD Level:										
Plains × Post-2007	-1.749*** (0.347)	-1.363*** (0.432)	0.251 (0.235)	0.624*** (0.173)	1.266 (1.642)	-0.077 (0.167)	-2.503*** (0.763)	0.980 (0.861)	0.425 (0.899)	-1.210 (1.057)
Dependent Variable Mean	3.393 (2.276)	3.579 (2.176)	1.381 (0.717)	1.509 (0.802)	-0.087 (1.853)	-0.340 (0.679)	-4.554 (3.948)	-3.846 (3.269)	13.780 (9.503)	13.330 (9.331)
Observations	624	624	598	598	624	624	623	623	624	624
Adjusted R^2	0.794	0.769	0.901	0.917	0.306	0.343	0.912	0.871	0.894	0.896
Adjusted Within R^2	0.107	0.067	0.049	0.233	0.027	0.001	0.161	0.028	0.001	0.007
Controls										
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	26	26	26	26	26	26	26	26	26	26

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The model compares the extreme PSD Plains to the Mountains. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize split-sample DiD framework and include County and Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE C6: ROBUSTNESS CHECK: TRUNCATED TIMELINE (2000–2019) WITHOUT CONTROLS

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Plains × Post-2007	-1.488*** (0.313)	0.266** (0.107)	0.600 (0.528)	-0.987* (0.533)	0.578 (0.769)
Dependent Variable Mean	3.033 (2.164)	1.606 (0.866)	0.042 (1.968)	-4.515 (3.704)	11.000 (6.214)
Observations	800	800	800	800	800
Adjusted R^2	0.828	0.935	0.343	0.911	0.782
Adjusted Within R^2	0.144	0.081	0.009	0.047	0.002
Controls					
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The sample timeline is truncated to 2000–2019 to test robustness against extreme years. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize baseline DiD framework and include County and Time Fixed Effects. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

D. Robustness

This section presents the synthetic falsification test introduced in [Section VI](#), as well as the formal estimates and visualizations for the sensitivity analyses outlined in [Section VII](#). [Tables D1, D2, D3, and D4](#) provide the regression outputs for the Low-PSD synthetic counterfactual, extreme TRI and PSD splits, as well as truncated timeline, respectively. [Figures D1, D2, D3, D4, and D5](#) display the leave-on-out spatial robustness checks for tractor density, rural business formation, rural natural increase, net migration rate, and fiscal dependency, respectively. Every figure gives the 40 coefficients with each county excluded at a time. Overall, all of the tests lend support to the core findings and suggest that they are structurally robust and not merely a result of arbitrary boundary definitions, early-transition volatility, or single-county outliers.

TABLE D1: SYNTHETIC DIFFERENCE-IN-DIFFERENCES RESULTS: LOW-PSD TEST

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
SDiD Estimate ($\hat{\tau}^{SDiD}$)	-1.037*** (0.305)	0.399** (0.189)	-0.060 (0.127)	0.005 (0.701)	-0.937 (0.878)
Observations	648	621	648	648	648
Treated Counties	6	6	6	6	6
Donor Pool (Mountains)	21	21	21	21	21
Time Weights ($\hat{\lambda}_t$)	Estimated	Estimated	Estimated	Estimated	Estimated
Unit Weights ($\hat{\omega}_c$)	Estimated	Estimated	Estimated	Estimated	Estimated
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Standard Errors	Jackknife	Jackknife	Jackknife	Jackknife	Jackknife

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The unit of observation is the county-year. Jackknife standard errors are reported in parentheses. All models include County and Time Fixed Effects. As a falsification test, this model matches the Low-PSD Plains sub-cohort to a synthetic Mountains baseline. Dependent variable means are omitted given that the SDID estimator evaluates effects relative to a weighted synthetic baseline rather than an unweighted sample average.

TABLE D2: ROBUSTNESS CHECK: EXTREME TRI SPLIT

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Extreme Plains $\times$ Post-2007	-2.368*** (0.539)	0.501*** (0.161)	-0.279 (0.167)	-0.439 (0.665)	-0.187 (1.371)
Dependent Variable Mean	2.893 (2.447)	1.748 (0.953)	-0.267 (0.674)	-5.449 (3.983)	14.000 (9.588)
Observations	460	460	460	459	460
Adjusted R^2	0.831	0.927	0.251	0.931	0.878
Adjusted Within R^2	0.299	0.251	0.051	0.113	0.078
Controls	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	20	20	20	20	20

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This sample restricts the dataset to the top 25% most rugged (Mountains) and bottom 25% flattest (Plains) counties. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize baseline DiD framework and include County and Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, Hospital Beds Index, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE D3: ROBUSTNESS CHECK: EXTREME PSD PLAINS SPLIT

Dependent Variable:	Tractor Density		Business Formation		External Migration		Rural Natural Increase		Fiscal Dependency	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)	High (7)	Low (8)	High (9)	Low (10)
Plains × Post-2007	-1.563*** (0.305)	-1.500** (0.545)	0.239 (0.242)	0.520*** (0.162)	1.567 (1.590)	-0.167 (0.173)	-2.322** (0.945)	0.716 (0.826)	0.398 (0.755)	-0.927 (0.981)
Dependent Variable Mean	3.393 (2.276)	3.579 (2.176)	1.381 (0.717)	1.509 (0.802)	-0.087 (1.853)	-0.340 (0.679)	-4.554 (3.948)	-3.846 (3.269)	13.780 (9.503)	13.330 (9.331)
Observations	598	598	598	598	598	598	597	597	598	598
Adjusted R^2	0.804	0.773	0.905	0.925	0.366	0.317	0.923	0.890	0.882	0.879
Adjusted Within R^2	0.143	0.078	0.097	0.309	0.112	0.015	0.245	0.134	0.071	0.045
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County	County	County	County	County	County
No. of Counties	26	26	26	26	26	26	26	26	26	26

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The model compares the extreme PSD Plains to the Mountains. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize split-sample DiD framework and include County and Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, Hospital Beds Index, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

TABLE D4: ROBUSTNESS CHECK: TRUNCATED TIMELINE (2000–2019)

Dependent Variable:	(1) Tractor Density	(2) Business Formation	(3) External Migration	(4) Rural Natural Increase	(5) Fiscal Dependency
Plains × Post-2007	-1.460*** (0.309)	0.273** (0.103)	0.716 (0.590)	-0.945* (0.547)	0.658 (0.707)
Dependent Variable Mean	3.033 (2.164)	1.606 (0.866)	0.042 (1.968)	-4.515 (3.704)	11.000 (6.214)
Observations	800	800	800	800	800
Adjusted R^2	0.831	0.939	0.364	0.915	0.793
Adjusted Within R^2	0.161	0.145	0.045	0.092	0.057
Controls	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Clustered SE	County	County	County	County	County
No. of Counties	40	40	40	40	40

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The sample timeline is truncated to 2000–2019 to test robustness against extreme years. The unit of observation is the county-year. Standard errors are clustered at the county level. The models utilize baseline DiD framework and include County and Time Fixed Effects. Controls include: Modernized Roads Ratio, High School Graduates Ratio, Hospital Beds Index, and Social Aid Index. Dependent variable means reflect the full estimation sample for each specification and differ across models due to differences in sample composition.

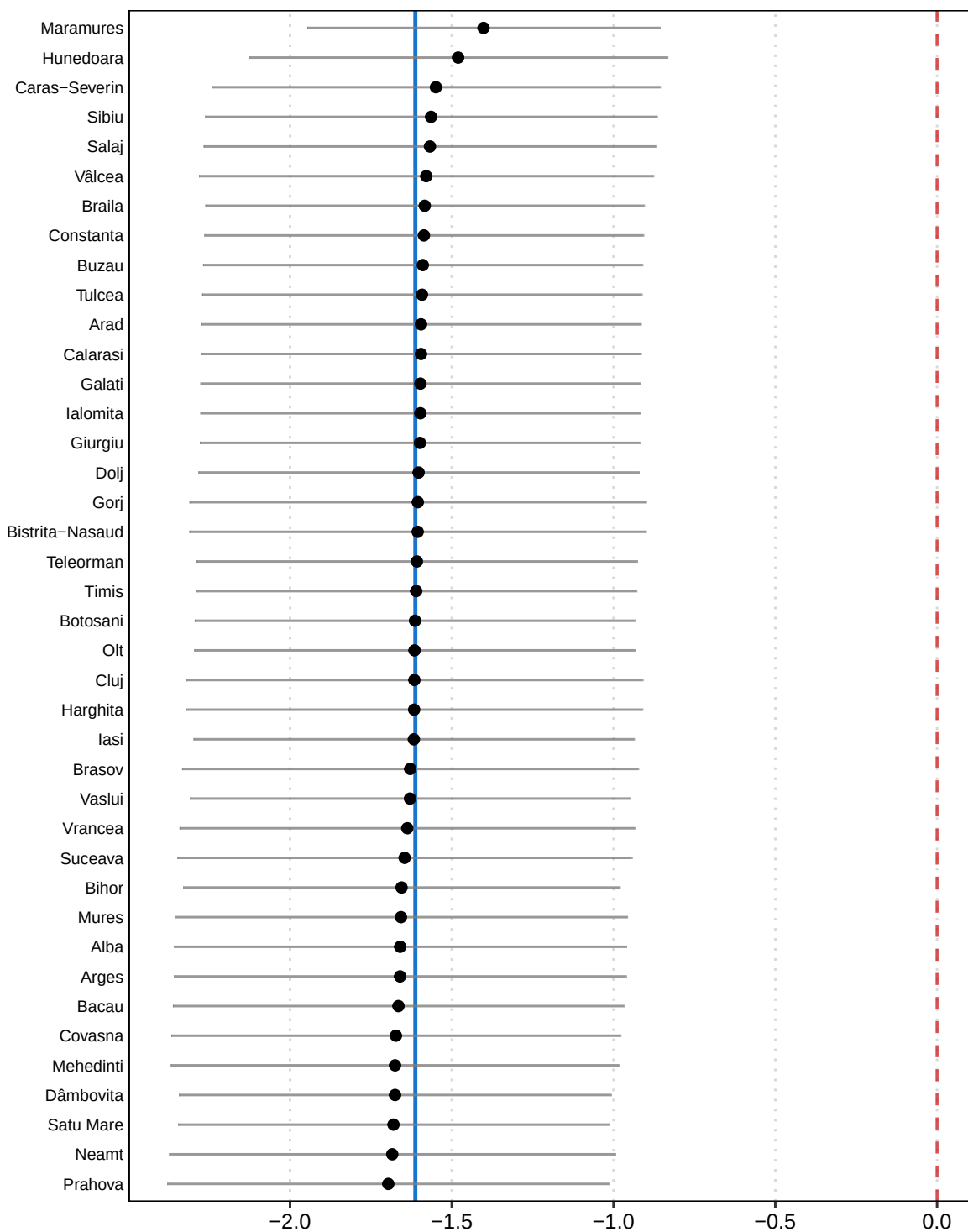


FIGURE D1: LEAVE-ONE-OUT ROBUSTNESS TEST FOR TRACTOR DENSITY

Notes: This figure presents point estimates and 95% confidence intervals for the baseline difference-in-differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

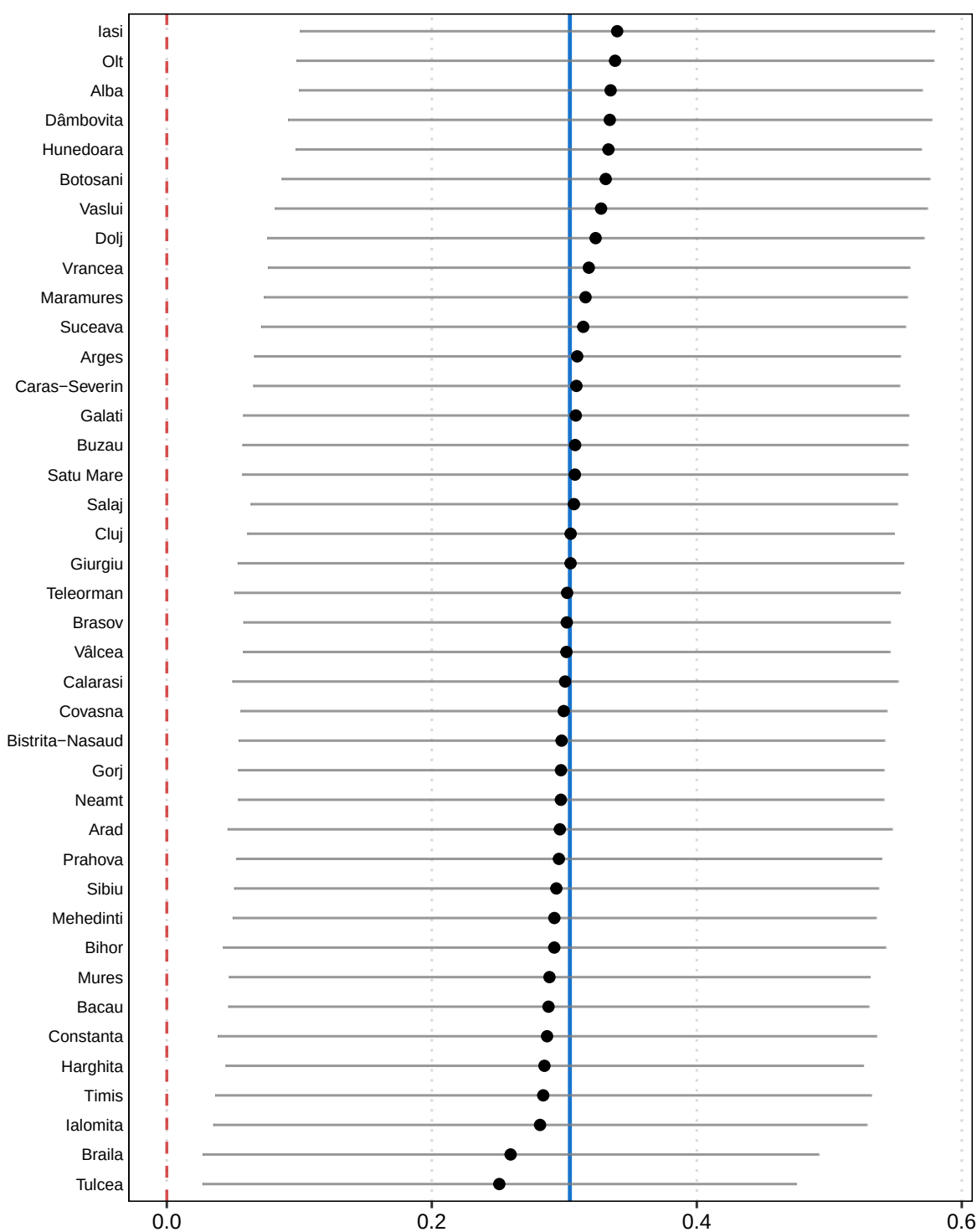


FIGURE D2: LEAVE-ONE-OUT ROBUSTNESS TEST FOR RURAL BUSINESS FORMATION

Notes: This figure presents point estimates and 95% confidence intervals for the baseline difference-in-differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

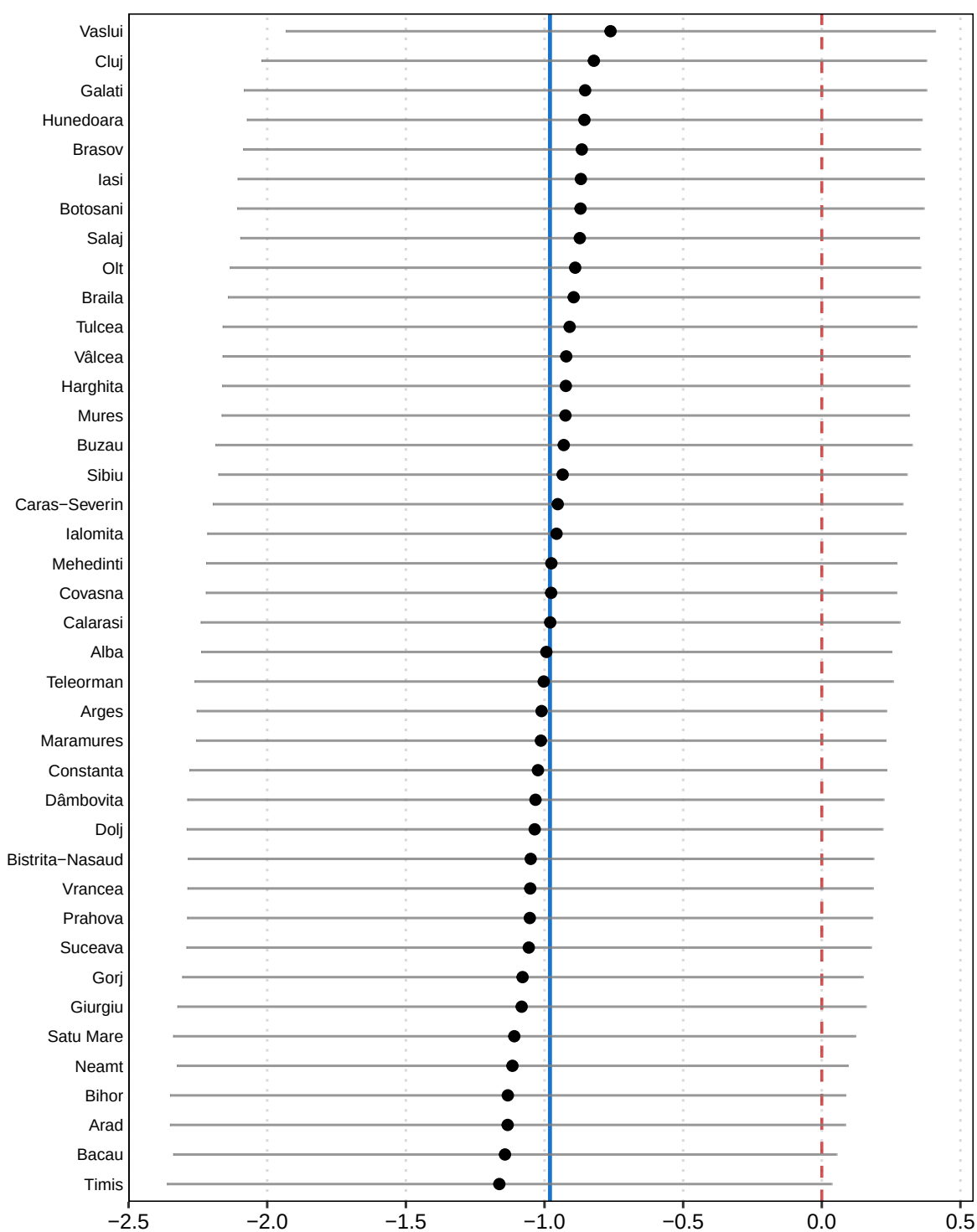


FIGURE D3: LEAVE-ONE-OUT ROBUSTNESS TEST FOR RURAL NATURAL INCREASE

Notes: This figure presents point estimates and 95% confidence intervals for the baseline difference-in-differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

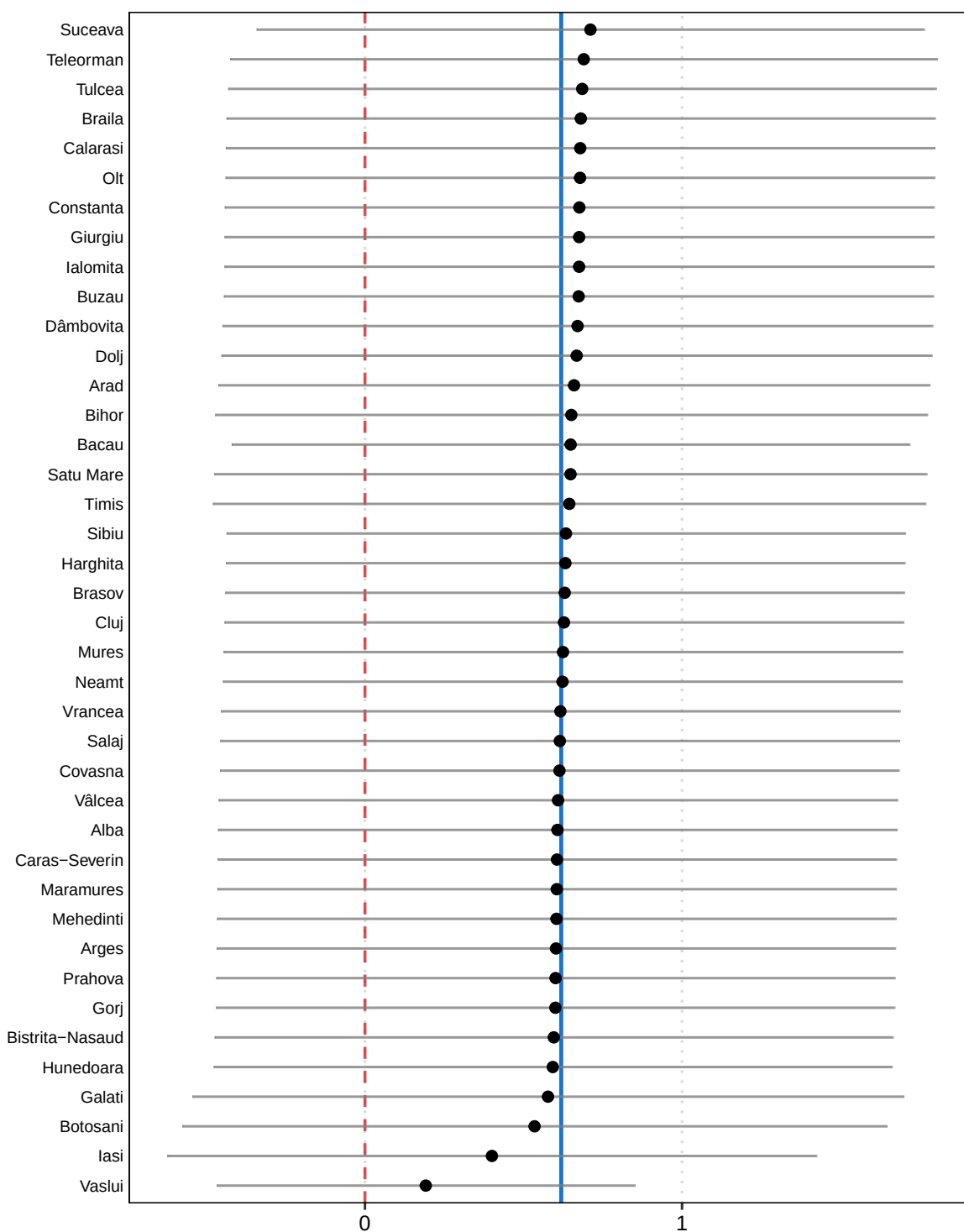


FIGURE D4: LEAVE-ONE-OUT ROBUSTNESS TEST FOR NET MIGRATION RATE

Notes: This figure presents point estimates and 95% confidence intervals for the baseline difference-in-differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.

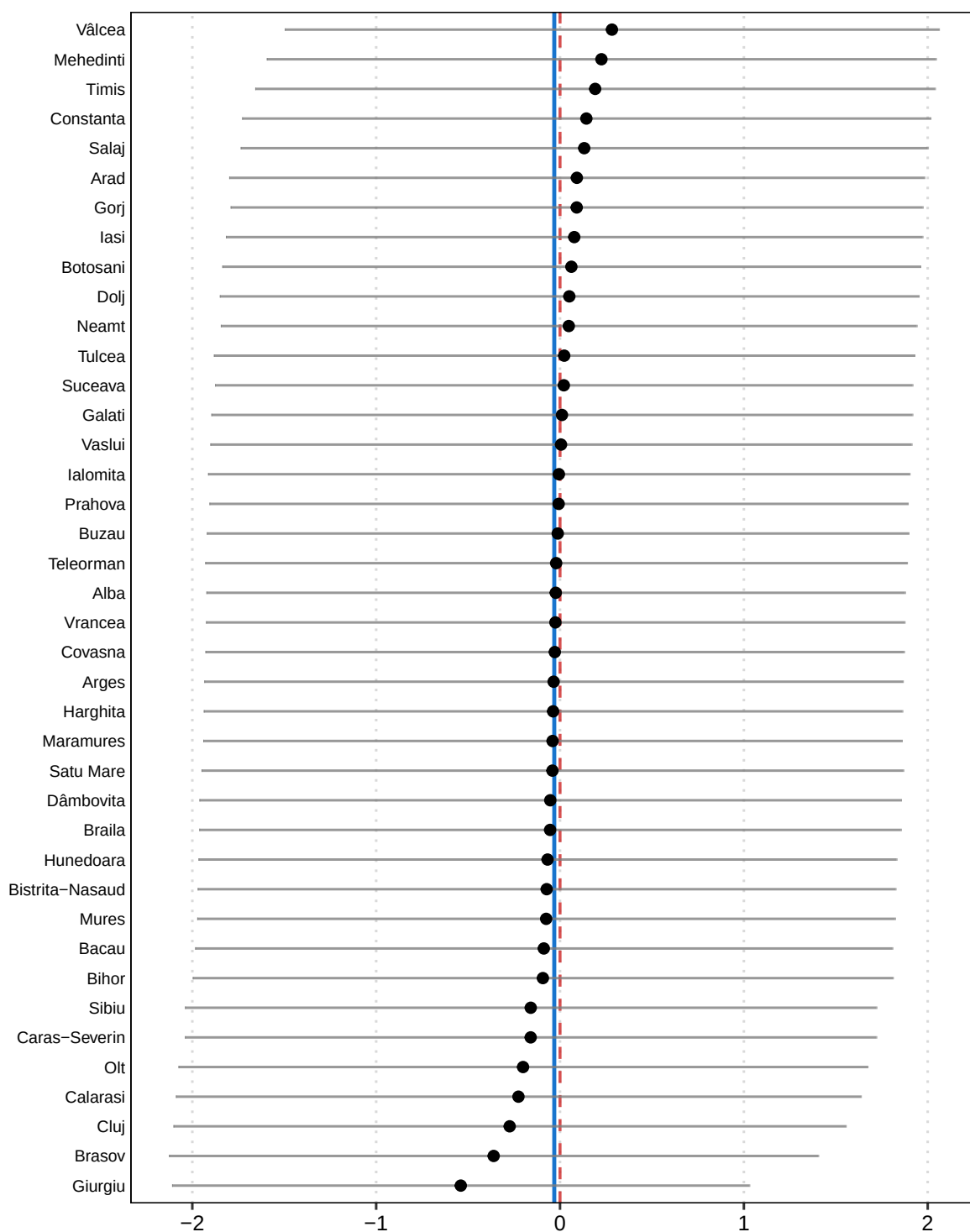


FIGURE D5: LEAVE-ONE-OUT ROBUSTNESS TEST FOR FISCAL DEPENDENCY

Notes: This figure presents point estimates and 95% confidence intervals for the baseline difference-in-differences coefficient when iteratively excluding the county listed on the y-axis. The solid vertical line represents the aggregate baseline estimate from the full sample, while the dashed red line indicates the zero threshold.